

Advancements in Obstacle Intrusion Detection Methods for Rail Transit: A Comprehensive Review

Xin-Yue Xu[✉], Su-Mei Wang[✉], *Member, IEEE*, Wen-Qiang Liu[✉], *Member, IEEE*,
and Yi-Qing Ni[✉], *Member, IEEE*

Abstract—Developing intelligent obstacle intrusion detection systems (OIDSs) is crucial for the safety of train operations. Advances in graphics processing units (GPUs) and image sensing technologies have driven research and development of railway environmental perception using artificial intelligence (AI). However, compared with road transport, the current research and development efforts of railway OIDS are relatively insufficient. Although several papers have reviewed obstacle detection methods in railway applications, a comprehensive summary and discussion of the state-of-the-art AI-based methods is lacking. In view of this, this article aims to provide a thorough review of the traditional visual sensor-based and AI-based obstacle intrusion detection (OID) methods used in railways. This article first discusses the characteristics and advantages of various sensing devices, including cameras, light detection and ranging (LiDAR), and millimeter-wave radar, as well as the integration of these sensors. The focus is on rail area detection algorithms and obstacle detection algorithms, and the algorithm principles and model structures are summarized. In addition, this article provides an in-depth discussion of strategies for improving and optimizing the OID algorithms. Finally, the challenges and future requirements of OIDS in railway applications are outlined, and potential research directions are highlighted.

Index Terms—Artificial intelligence (AI), computer vision, obstacle intrusion detection (OID), rail transit.

I. INTRODUCTION

High-speed railways, subways, and tramways are widely constructed globally, serving as the primary modes of transportation for both passengers and freight due to their cost-effectiveness, speed, and safety. However, rail accidents remain a significant problem, mainly due to obstacles intruding onto the tracks. According to the International Union of Railways (UIC), 90% of railway accidents are attributed to third-party intrusions into rail lines [1]. Given the limited reaction time and field of vision of train drivers, it is difficult to avoid

collisions with obstacles through human observation alone. These intrusions pose a serious risk to train safety and require urgent attention.

Traditional manual inspection methods are gradually being replaced due to their low efficiency, high cost, and strong subjectivity. Sensing devices, together with image processing and artificial intelligence (AI) techniques, have been used to develop an obstacle intrusion detection system (OIDS) for railway applications [2], [3], [4], [5], [6], [7], [8], [9]. High-resolution visual sensors, such as cameras, light detection and ranging (LiDAR), and millimeter-wave radar, are typically installed on locomotives, level crossings (LCs), and stations. These sensors [271] in conjunction with obstacle intrusion detection (OID) algorithms to detect obstacles that are currently or potentially entering the rail area, thereby alerting the running trains of the danger. For instance, Ristić-Durrant et al. [10] presented an onboard multisensor system that includes a night vision sensor, three RGB cameras, and a 3-D laser scanner as part of the H2020 Shift2Rail project SMART. They utilized an object detection method to identify obstacles and constructed a multilayer neural network to estimate the distance from the obstacle to the vehicle. He et al. [11] developed a front-end acquisition system that incorporates a LiDAR, an infrared (IR) camera, a visible light camera, and a millimeter-wave radar. They applied the Mask R-CNN network to detect rail areas and obstacles simultaneously.

OIDS methods can be broadly divided into two approaches. The first approach is difference analysis, which detects obstacles by comparing pixel or image differences using modeling or reconstruction techniques. The second approach consists of two components: rail detection and obstacle detection. In this approach, rail regions are first identified through trajectory detection or track segmentation, followed by the identification of obstacles within these regions using object detection and segmentation techniques.

The algorithms used in OID can be categorized into traditional image processing techniques and AI-based deep learning (DL) methods. Traditional image processing relies on local operations to extract shallow, low-level features from rail and obstacle images. For instance, traditional object detection often adopts sliding window methods and manually designed features [e.g., Haar, scale-invariant feature transform (SIFT), and histogram of oriented gradients (HOG)] to match targets based on feature similarity. However, the complexity of manual feature design, coupled with the limitations of sliding window

Received 14 March 2025; revised 13 August 2025; accepted 25 August 2025. Date of publication 22 September 2025; date of current version 8 October 2025. This work was supported in part by Wuyi University's Hong Kong and Macao Joint Research and Development Fund under Grant 2019WGALH15 and Grant 2019WGALH17; in part by the Innovation and Technology Commission of Hong Kong SAR Government, China, under Grant K-BBY1; and in part by the Research Grants Council (RGC) of the Hong Kong Special Administrative Region (SAR), China under Grant PolyU 152308/22E. The Associate Editor coordinating the review process was Dr. Jianbo Yu. (*Corresponding author: Yi-Qing Ni.*)

The authors are with the Department of Civil and Environmental Engineering and the National Rail Transit Electrification and Automation Engineering Technology Research Center (Hong Kong Branch), The Hong Kong Polytechnic University, Hong Kong, China (e-mail: xinyue-cee.xu@connect.polyu.hk; may.sm.wang@polyu.edu.hk; wqiang.liu@polyu.edu.hk; ceyqni@polyu.edu.hk).

Digital Object Identifier 10.1109/TIM.2025.3612624

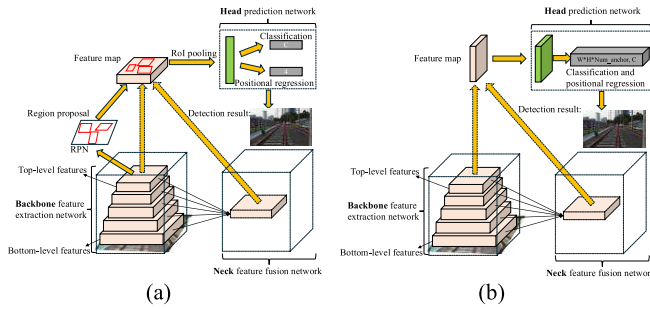


Fig. 1. Object detection frameworks based on DL. (a) Two-stage object detection framework. (b) One-stage object detection framework.

size and search methods, makes this traditional approach inefficient for OID. Traditional segmentation methods classify regions based on inherent object properties (e.g., color, texture, and spatial contours) and cluster these regions using manually designed clustering strategies (e.g., edge detection, thresholding, region growing, graph partitioning, and shape priors). While computationally efficient, these methods rely heavily on qualitative descriptions or scene-specific assumptions, making them less adaptable to different rail environments, such as railway turnouts, crossings, line gaps, and obscured tracks.

In contrast, DL-based methods, such as convolutional neural networks (CNNs), can automatically extract features with high abstraction, accuracy, and robustness and are therefore widely used in OID. Given the limitations of computational resources and the high-performance demands for railway applications, current research has focused on designing efficient and accurate object detection and segmentation frameworks to efficiently abstract and quantify image content. The current mainstream CNN-based object detection frameworks (see Fig. 1) are divided into two categories: two-stage object detection networks based on region proposals (RPs, e.g., the R-CNN series [12], [13], [14]) and one-stage object detection networks (e.g., OverFeat [15], SSD [16], and YOLO [17] series). For CNN-based segmentation networks, two main types dominate (see Fig. 2): forward-structured segmentation networks [18], which directly predict pixel categories in the output layer, and encoder-decoder structured segmentation networks [19], which predict pixel categories by abstracting and reconstructing features with the same resolution as the image (e.g., SegNet [20] and U-Net [21]). As illustrated in Figs. 1 and 2, these frameworks typically consist of three subnetworks: backbone network, neck network, and head network. The backbone network acts as the primary feature extractor, processing images or videos to generate feature maps [22], [23], with accuracy, speed, and efficiency being the key factors in its selection. The neck network fuses different feature maps, while the head network is responsible for making the final prediction. Improvements to OID focus on enhancing backbone feature extraction methods (e.g., VGG [24] and ResNet [25]) and neck feature fusion methods (e.g., feature pyramid network (FPN) [26] and PANet [27]). In the segmentation network, the forward-structured network focuses on integrating low-level texture features with high-level semantic features, while the encoder-decoder network emphasizes fusing information

across different feature layers, combining high-level semantic features with low-level detailed features.

In addition to the optimization and refinement of each subnetwork, multiscale feature fusion, effective attention mechanisms, lightweight network design, and specialized loss functions are also adopted into the DL network to further enhance robustness in complex railway scenarios. Multiscale feature fusion improves network performance by combining features across different resolutions. U-Net-style skip connections enable shallow and deep feature merging, providing auxiliary semantic and spatial information for the final prediction. However, this fusion is often limited to the same-resolution feature matching, hindering the use of diverse and complementary information. Advanced methods such as FPN [26], PANet [27], BiFPN [28], and SSPNet [29] address this by building hierarchical pyramids, bidirectional fusion paths, learnable fusion weights, and shareable features to enhance richer cross-scale interaction. In addition, techniques such as StairNet [30] enhance spatial detail recovery via learnable upsampling. IPG-Net [31] uses an IPG conversion module to recover fine details from shallow features. Methods such as Auto-FPN [32], NASFPN [33], and EfficientDet [28] leverage neural architecture search (NAS) to optimize fusion paths dynamically. Multiscale feature fusion technology is evolving toward structured, learnable, and dynamic fusion mechanisms, which are especially beneficial in complex railway object detection scenarios with varying object sizes and occlusions. Attention mechanisms, including spatial, channel, and self-attention, have proven effective in refining feature representations. Spatial attention applies masks to optimize spatial features, guiding the model to focus on regions of interest (ROIs) by pixel-level arrangements, while channel attention recalibrates feature importance by selectively weighing feature channels. Their combinations (e.g., convolutional block attention module (CBAM) [34]) improve focus on relevant targets. Self-attention, popularized in Transformer, captures long-range dependencies across entire images by computing attention scores for each pixel to all other pixels within the input feature map. Advanced variants such as EMANet [35], CCNet [36], and BiFormer [37] further enhance performance and efficiency. Emerging approaches like probabilistic attention [38], [39], [40] further refine performance by concentrating on sparse but informative regions. Attention mechanisms have evolved from heuristic designs to more sophisticated designs based on statistical principles, particularly valuable in cluttered or visually challenging railway environments. In addition, to meet real-time constraints and enable deployment on edge devices, lightweight network designs have been proposed using depthwise separable convolutions, convolutions with channel shuffling, bottleneck structures, simplified attention mechanisms, and other efficient modules to reduce computational cost. Meanwhile, tailored loss functions such as focal loss, CIoU loss, Dice loss, and hybrid loss are often introduced to address class imbalance and enhance localization precision in OID. Furthermore, low-level image enhancement techniques, such as denoising, deblurring, defogging, deraining, and super-resolution, are applied to mitigate environmental effects and improve visual quality. Section IV will discuss

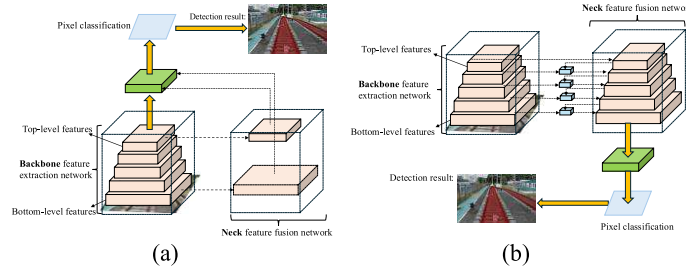


Fig. 2. Object segmentation framework based on DL. (a) Forward-structured segmentation framework. (b) Encoder-decoder structured segmentation framework.

the OID improvement strategies based on these subnetworks reported in the literature.

Although several papers reviewed the use of OIDS in railway applications, many papers focused only on pedestrian detection methods developed for road vehicles or on traditional computer vision techniques for rail track and obstacle detection. A comprehensive and systematic summary of the state-of-the-art AI-based methods is lacking. A thorough understanding of AI algorithms applied to OID in railways is crucial to improving the operational safety of rail transit. Therefore, this article summarizes the latest advancements and practices in intelligent OIDS in terms of sensing devices, railway scene datasets, and traditional and state-of-the-art methods for rail and obstacle detection. This review provides a valuable resource for researchers in the railway field and aims to fill the gaps in previous research. The main contributions of this work are as follows.

- 1) A detailed introduction to the sensor schemes used in OIDS for railway applications, along with currently available benchmark datasets for railway scenes and obstacles.
- 2) A methodological review and summary of traditional image processing-based and AI-based OID methods, including a discussion of improvement strategies for models applied in railway OIDS.
- 3) An analysis of the challenges faced by current OID methods in practical applications and potential solutions.

The remainder of this article is organized as follows. Section II introduces the principles and advantages of three types of sensors used for OIDS, as well as their combination schemes. Section III presents various railway benchmark datasets and their annotation methods. Section IV summarizes and discusses traditional and state-of-the-art AI-based detection algorithms for OID in railway applications, as well as strategies to improve these methods. Section V outlines the challenges faced by OID algorithms in practice and explores potential solutions and future directions for breakthroughs. Finally, conclusions and prospects are presented in Section VI.

II. SENSORS FOR DATA ACQUISITION

Effective sensors play a vital role in the development of OIDS. Some sensors can even exceed human perception: camera telephoto lenses extend the distance of view, while Radar and IR cameras perform better in adverse weather conditions such as fog, rain, or snow. LiDAR and IR technologies are particularly effective for obstacle recognition in

low-light or night-time conditions. Currently, railway OIDS mainly relies on cameras, LiDAR, and millimeter-wave radar. These sensors can be mounted on locomotives, LCs, and stations to capture images, point clouds, and electromagnetic wave data of obstacles that intrude into rail areas [3], [7], [41], [42], [43], [44], [45], [46], [47], [48], [49], [50], [51].

Cameras are a passive measurement scheme that captures images using light reflected from an object's surface, projecting its physical features (such as color, contour, and texture) onto an optical sensing chip with varying resolutions. The data obtained by visible light cameras are dense and rich in visual information, making them effective for OIDS to identify object boundaries and classify obstacles [4]. However, these cameras have limited functionality in low-light conditions. In contrast, IR cameras detect IR radiation and capture IR light reflected from surrounding objects, enabling night vision capabilities. While IR cameras are straightforward to operate and require minimal power, they have lower accuracy in tracking and positioning.

LiDAR is an active measurement scheme that uses laser signals of specific wavelengths to scan the surface of an object and collect spatial information. It calculates the distance to the obstacle by calculating the travel time of the light wave based on the received reflected signal. LiDAR has strong anti-interference ability, is not affected by changes in ambient light, has excellent directionality and coherence, and can achieve high-precision distance measurement. Compared to other sensors, LiDAR excels in obstacle detection and environmental reconstruction, but its detection range is relatively short, and it performs poorly in rainy or foggy conditions.

Millimeter-wave radar is a type of radar that operates in the 1–10-mm electromagnetic waveband. Like LiDAR, millimeter-wave radar is an active measurement scheme. However, instead of emitting linear electromagnetic waves with concentrated energy, it emits electromagnetic waves in a conical beam and measures the distance, relative speed, and direction of an object based on the frequency difference between the transmitted and received signals. Millimeter-wave radar is not affected by ambient light and airborne obstacles and has advantages under various conditions. However, its hardware is relatively complex and costly, and the measurement results may be rough and noisy.

The above three types of sensing devices can be carried on different operating carriers (trains, drones, and satellites). As shown in Fig. 3, satellites usually carry cameras to capture remote sensing images from space [52], [53], and trains

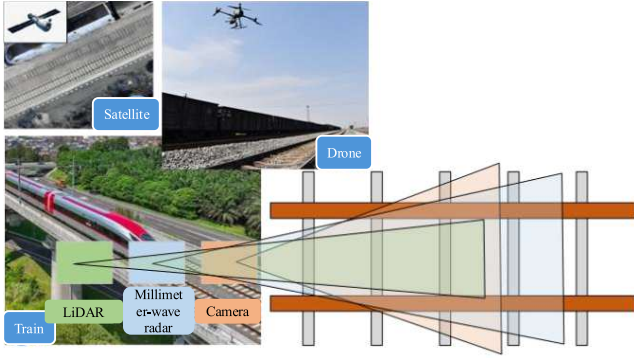


Fig. 3. OIDS solutions.

TABLE I

COMPARISON OF CAMERA, LiDAR, AND MILLIMETER-WAVE RADAR SENSORS IN THE DETECTABLE DISTANCE RANGE

Performance	Camera	LiDAR	Radar
Resolution	Camera > LiDAR > Radar		
Weather resistance	Camera & LiDAR < Radar		
Illumination resistance	Camera < LiDAR & Radar		
Track target speed	Camera & LiDAR < Radar		
Track target height	LiDAR > Camera > Radar		
Track target distance	Camera < LiDAR & Radar		
Targets recognition	Camera & LiDAR > Radar		
Cost	Camera > LiDAR > Radar		

TABLE II

SUMMARY OF MULTISENSOR SCHEMES

Sensors	References	Details
camera & laser	[57]	an <u>infrared camera</u> and a <u>laser range finder</u>
	[58]	an 8-layer <u>laser scanner</u> , an <u>RGB camera</u> , and a <u>thermal camera</u>
	[59]	a <u>camera</u> and a <u>near-infrared laser</u>
camera & radar	[60], [61]	a <u>radar</u> and a <u>camera</u>
	[62]	three <u>cameras</u> and an <u>infrared radar</u>
	[63], [64]	a <u>camera</u> and a <u>millimeter-wave radar</u>
camera & laser & radar	[65]	<u>stereo vision</u> , <u>mono cameras</u> , a <u>radar</u> , and a <u>laser</u>
camera & LiDAR	[66]	one short-range and two long-range <u>cameras</u> and <u>LiDAR</u>
combination cameras	[59], [67]–[81]	<u>RGB cameras</u> (including short or long-range cameras, grayscale cameras, thermal cameras, and infrared cameras)
stereo cameras	[65], [82]	two or three <u>cameras</u>

and drones can carry all three sensors at the same time [54], [55], [56]. The data acquisition principles of the three sensors are different, and each of them has unique advantages and specific limitations. Table I compares the performance of cameras, LiDAR, and millimeter-wave radar sensors in terms of resolution, weather resistance, illumination resistance, ability to detect track target speed, height and distance, target recognition capability, and cost within the detectable range.

The choice of specific sensors depends on factors such as obstacle detection environment, operating conditions, and

costs. Given that each sensor has its own advantages and limitations, sensor fusion is a common solution for OIDS. For example, LiDAR–camera fusion, radar–camera fusion, and laser–camera fusion schemes are adopted to enhance the depth prediction performance of detection methods. Using stereo cameras with better angular resolution than monocular cameras can provide more accurate obstacle location information, thus facilitating trajectory and ego-motion estimation. In addition, thermal or IR camera schemes can be used at night without relying on external light sources, thereby reducing system energy consumption. Recent studies on multisensor schemes [46], [57], [58], [59], [60], [61], [62], [63], [64], [65], [66], [67], [68], [69], [70], [71], [72], [73], [74], [75], [76], [77], [78], [79], [80], [81], [82] are summarized in Table II. Considering the object recognition capability, current OIDS primarily focuses on the data captured from camera and LiDAR sensors.

III. BENCHMARK DATASETS

Constructing a railway benchmark dataset is essential for studying OIDS detection algorithms and verifying their timeliness, robustness, and applicability. One of the most direct ways to build a railway benchmark dataset is to install sensors on operating trains to collect data and then annotate them manually. However, this process is often very complicated due to practical constraints such as the need for appropriate qualifications to obtain onboard data, the difficulty of collecting on-site obstacle data, and the limited variety of obstacles encountered. Therefore, researchers have integrated a large number of railway scene images from the Internet or adopted virtual data synthesis techniques to generate effective obstacle images and then manually or automatically annotated the images to construct a railway benchmark dataset. This section summarizes several established rail and obstacle detection benchmark datasets and their labeling rules, as shown in Table III.

A. On-Site Railway Scene Datasets

Zendel et al. [83] launched RailSem19, the first open-source dataset built from the ego-vehicle perspective for semantic scene understanding in train and tram driving. This dataset comprises 8500 images captured from the driving view and includes short annotated sequences with over 1000 railway crossing examples and 1200 tram scenes. RailSem19 provides dense pixel-level semantic labels that map Cityscapes training labels to RailSem19 dense labels, along with color legends, and includes manual annotations in the form of geometric shapes. Toprak et al. [84] introduced RAWPED, a challenging benchmark designed for developing and evaluating pedestrian detection methods for onboard driver assistance systems. The dataset comprises 26 000 annotated frames from the ADORAS project and public sources, capturing real-world railway conditions with varying resolutions, occlusions, and pedestrian appearances across different railway environments. Hadded et al. [85] constructed the second large-scale rail segmentation dataset, RailSet, based on driver views of trains under various conditions, climates, and acquisition periods.

TABLE III
BENCHMARK DATASETS AND ANNOTATION LEVELS

Type of dataset	Dataset	Ref.	Sensors	Resolution/ Data size	Frame Rate/ Freq.	Annotation level
On-site railway scene datasets	RailSem19	[83]	RGB camera	variable	-	pixel-level and object-level
	RAWPED	[84]	RGB video	variable	25-30fps (<20km/h); 50fps (100km/h)	object-level
	RailSet_Hadded	[85]	RGB video	variable	-	pixel-level
	RailEye3D	[86]	Stereo cameras	1088×1920 pixels	-	object-level
	OSDaR23	[87], [88]	LiDAR	50k–84k pts/frame (long); 60k–115.2k pts/frame (medium); 20k–40k pts/frame (short)	10Hz	pixel-level and object-level
			2D Radar	2856×1428 pixels	4Hz	
			RGB camera	4112×2504 pixels	10Hz	
			IR camera	640×480 pixels	10Hz	
			HD camera	1920×1080 pixels	20fps	
	MRSI	[89]	Black light camera	1920×1080 pixels	-	pixel-level and object-level
			IR camera	640×512 pixels	-	
	Rail-DB	[90]	RGB video	3840×2160 pixels; 1280×720 pixels; 1920×1080 pixels	-	pixel-level
	WHU-Railway3D	[91]	LiDAR	4614M pts	-	pixel-level
	Rail3D	[92]	LiDAR	288M pts	-	pixel-level
	RailGoerl24	[93]	RGB camera	1920×1080 pixels	-	object-level
			LiDAR	384M pts	-	
	Gebler <i>et al.</i>	[94]	Dynamic vision sensor	360×640×2 pixels (Time Surface); 360×640×6 pixels (Event Volume)	-	image-level
Synthetic obstacle datasets	Franke, M. <i>et al.</i>	[95]	RGB	1024×1024 pixels/ 2026 frames	-	object-level
	RailSet_Zouaoui	[96]	RGB camera	256×256 pixels	-	pixel-level
	SynRailObs	[97]	RGB video	1280×720 pixels	-	object-level
	Ganter, J. <i>et al.</i>	[98]	RGB camera	1600×1200 pixels/ 10291 frames	-	pixel-level
Virtual modeling datasets	Sevi, M. <i>et al.</i>	[99]	RGB camera	3840×2160 pixels	10-30 Hz	pixel-level
	TrainSim	[100]	LiDAR	-	-	pixel-level
			RGB camera	1088×1920 pixels	-	
	SOSDaR24	[101]	LiDAR RGB camera	- 1232×1028 pixels	-	pixel-level and object-level

RailSet focuses on labeling railroads and their classification, integrating railroad data from various natural environmental factors, and bridging the gap between road and rail domains in terms of datasets and experiments. Wallner et al. [86] introduced RailEye3D, a stereo vision (RGB-D) dataset collected using binocular cameras for train platform safety applications. It includes depth data and annotated images of people and objects such as wheelchairs and backpacks. Annotations cover tracking IDs, occlusion levels, and safety zones, supporting detection and multiobject tracking tasks.

In addition to rail datasets built using cameras, other sensor devices have also been used to collect railway visual data. Tagiew et al. [87], [88] created an open-source dataset, OSDaR23. The data were collected in Hamburg, Germany, and consists of 21 calibrated and synchronized IR and color (RGB) cameras, LiDAR, acoustic radar, and accelerometers mounted on locomotives. In addition to the raw data, it also contains 204 091 annotations, including folds, polygons, and rectangles, covering 20 different object categories to meet the needs of

various detection scenarios. Chen et al. [89] constructed the first multimodal remote sensing dataset, MRSI. This dataset comprises 27 000 images collected from freight railways and subways in Guangdong, China, and all objects are manually labeled using pixel and box annotations. It captures railway scenes, including straight paths, curves, and sharp turns, in a variety of lighting and weather conditions, including daytime, dusk, nighttime, and rainy days. In addition, an IR thermometer was integrated into the subway environment, and a new image alignment method was proposed by combining the spatial and radiometric properties of simultaneous acquisition. Also, Li and Peng [90] introduced a railway dataset called Rail-DB, which contains 7432 pairs of images and annotations. The images were collected under different lighting conditions, railway structures, and view scenarios. The railway tracks are annotated with polylines, and the dataset is divided into nine scenes. To promote point cloud semantic segmentation in railway scenarios, Qiu et al. [91] established WHU-Railway3D, a large-scale dataset for railway scene understanding. It contains

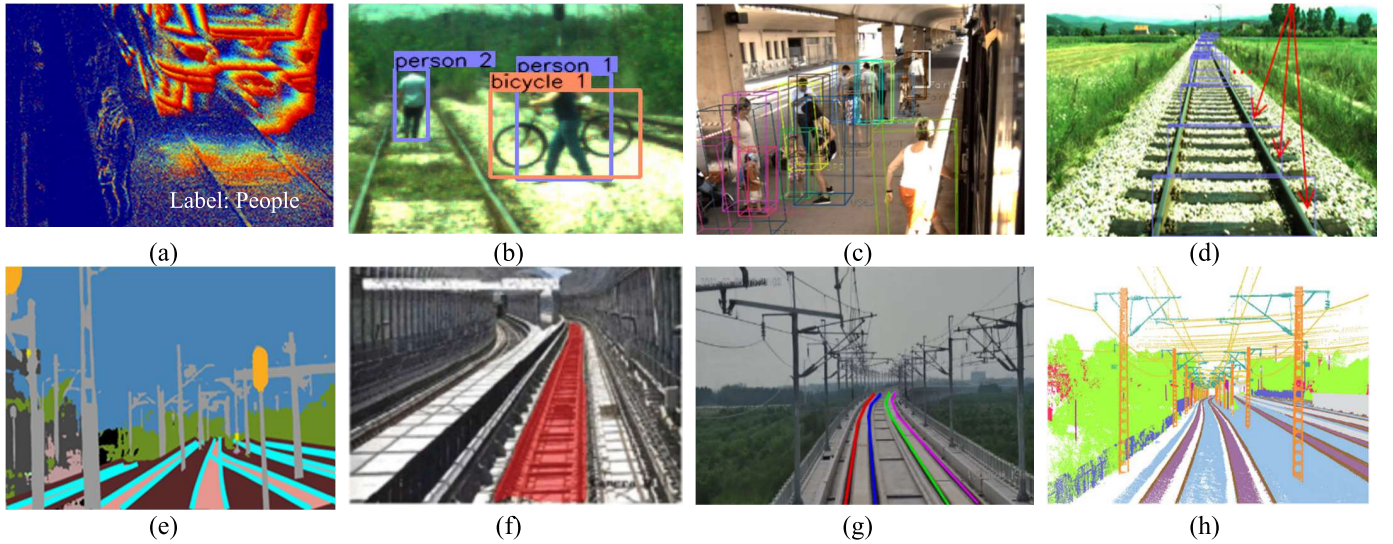


Fig. 4. Examples of images with different annotation levels. (a) Image-level annotation [94]. (b)–(d) Object-level annotation [86], [95]. (e)–(h) Pixel-level annotation [68], [83], [90], [91].

over 4.6 billion points collected along 30 km of railways, covering urban, rural, and plateau environments. The data were acquired using mobile laser scanning systems equipped with global navigation satellite system (GNSS), IMU, LiDAR, and cameras. The dataset includes 11 labeled object categories and serves as a benchmark for evaluating DL methods in complex railway environments. Kharroubi et al. [92] proposed Rail3D, a multicontext 3-D point cloud dataset containing railway scenes collected from Hungary, France, and Belgium. It includes various railway types, track conditions, and scene densities. The dataset provides nine key annotated semantic classes to support railway scene understanding tasks and includes multiple environments to reflect different operational settings. Tagiew et al. [93] introduced RailGoerl24, a dataset collected from a railway test center in Görlitz, Germany, designed to support research on driverless trains. It includes 12 205 RGB image frames with 33 556 bounding box annotations for persons, focusing on obstacle detection in railway environments. Part of the area is also covered by unannotated 3-D point cloud data. The dataset applies to visual perception tasks in urban rail transit and trunk railways. Gebler and Pohle-Fröhlich [94] built a dataset for railway tunnel entrance intrusion detection based on event camera, collecting 11 weeks of event streams at the tunnel and campus-simulated pedestrian activities. It includes natural scenes (trains, rain/snow, and lighting noise) and composite samples (merged event streams for risky scenarios), labeled as “People” and “NoPeople.”

B. Synthetic Obstacle Datasets

Unlike rail scene datasets, the primary challenge in establishing railway scene obstacle intrusion datasets is the lack of obstacle data and the difficulty in obtaining a large amount of on-site obstacle data. Consequently, data synthesis and data generation have emerged as feasible solutions.

Franke et al. [95] proposed an obstacle data fusion method. This approach involves fusing object-level annotated images

(e.g., pedestrians, animals, and fallen rocks) from open-source datasets into railway regions from other images. Various image preprocessing techniques were employed to ensure that the image style in the target domain remained as consistent as possible with that of the source domain, thereby enhancing the authenticity of the synthesized images. Zouaoui et al. [96] introduced RailSet, an innovative dataset for detecting railway anomalies. This dataset comprises 6600 high-quality manually annotated images depicting normal railway conditions, along with 1100 images of railway defects such as hole anomalies and rail discontinuities. They used the StyleMapGAN generative DL algorithm to generate images of abnormal scenarios. Guo and Rambach [97] constructed SynRailObs, a synthetic dataset for railway obstacle detection, combining real-world backgrounds with foreground obstacles using stable diffusion. It covers diverse environments and uses auxiliary datasets and SAM-based blending to create realistic scenes.

C. Virtual Modeling Datasets

Building an OID dataset based on on-site railway data often fails to meet the requirements of diverse scenes and object categories. In order to simulate various scenarios in the natural environment and enrich the diversity of data samples, the use of virtual reality environments together with 3-D modeling and rendering engines is an effective solution for automatically creating and annotating railway data using software algorithms.

Ganter et al. [98] proposed a virtual dataset with semantic enhancement and automatically generated annotations. Sevi and Aydın [99] developed a virtual rail inspection environment using drones in the Gazebo simulation environment, where they segmented the rail using a two-stage segmentation network. D’Amico et al. [100] proposed a visual simulation framework named TrainSim, which enables the generation of realistic railway scenes in virtual environments. It can automatically generate various labeled datasets from various

sensors, including video cameras, LiDAR, and IMUs. Users can define sensor positions and environmental settings to simulate realistic operating conditions. These scenes and datasets can be used to train innovative algorithms for tasks such as segmentation, object detection, distance estimation, and trajectory detection. Minaeian et al. [101] introduced synthetic open sensor data (SOSDaR24) to address the scarcity of publicly accessible datasets. The dataset was generated using the CARLA simulator to mitigate the costs and risks associated with real-world railway data collection. SOSDaR24 contains 100 different scenes and provides 22 208 files for each sensor, capturing both static and moving obstacles such as boxes, pedestrians, and vehicles. The dataset is annotated according to the OpenLABEL standard, providing details such as 3-D rail polylines, obstacle bounding boxes, and object class labels.

D. Annotation Levels

The levels of annotation (image level, object level, and pixel level) directly determine the type of computer vision-based modeling scheme and detection results. In OSDaR23 [87], [88], SOSDaR24 [101], and MRSI [89], railroad tracks and objects are labeled with both pixel-level semantic and object-level annotations. RailSem19 [83] includes pixel-level semantic annotations and object-level annotations of railway scenes, such as rail-related polygons and polylines. RailDB [90], RailSet [85], [96], and datasets constructed by Ganter et al. [98], Sevi and Aydın [99], and TrainSim [100] provide dense pixel-level semantic annotations of rail tracks. In contrast, the datasets built by Franke et al. [95], RAWPED [84], RailEye3D [86], RailGoerl24 [93], and SynRailObs [97] are annotated with object-level annotations. WHU-Railway3D [91] and Rail3D [92] include 3-D semantic annotations. Furthermore, the dataset built by Gebler and Pohle-Fröhlich [94] frames the intrusion problem as an image classification task by labeling “People” and “NoPeople” to clips. Segmentation models are typically used for pixel-level annotations, while object detection models commonly use object-level annotations. Object-level annotations can usually reduce labeling costs, while pixel-level annotations provide more precise location and shape information about the target, and richer interior and boundary details. Different models have their own advantages and applicability when dealing with different labels and tasks. Fig. 4 gives the examples of images with different annotation levels.

IV. RAILWAY OID METHODS

This section provides an overview of computer vision-based OID methods from a methodological perspective and critically evaluates typical algorithms, technical features, advantages, limitations, and improvement strategies proposed in the literature. The current OID methods can be broadly divided into two approaches. The first approach is difference analysis, which detects obstacles by comparing the differences between pixels or images using modeling or reconstruction techniques. This approach is discussed in Sections IV-A and IV-B3. The second approach involves two parts: rail detection and obstacle detection. First, rail regions are extracted through trajectory

detection or track segmentation, and then, obstacles within these regions are identified using object detection and segmentation methods. This approach is presented in Sections IV-B1, IV-B2, IV-B4, and IV-C. From a methodological perspective, OID algorithms based on these two approaches can be categorized into traditional image processing techniques and AI-based DL methods. In addition, track trajectory detection methods (Section IV-C), obstacle distance estimation methods (Section IV-D), and sensor fusion and event-based vision methods (Section IV-E) are also applied in OIDS. The details of these methods are summarized in Tables IV and V.

A. Traditional Image Processing Methods for OID

The distance between the rail tracks and other geometric details of the railway is a priori known in each country. In the bird’s-eye view (BEV), the geometry of the railroad track consists of two parallel lines whose slopes change gradually over short distance. In the projected image, the rail lines converge and intersect at a vanishing point in the distance. This perspective projection also alters the perceived scale of detected objects, making nearby objects appear larger than distant objects, even though their actual scale is the same. Based on these characteristics, some researchers have proposed rail and obstacle detection algorithms, which can be divided into projected image-based methods, BEV-based methods, and consecutive frame-based methods. For rail detection, traditional image processing methods primarily utilize local operations to extract low-level image features such as grayscale, color, texture, and edges of rail tracks to generate suitable initial candidates. The candidate rail pixels are subsequently fit to straight line segments using techniques such as multiscale edge extraction using 2-D Gabor wavelets [124] and merging preliminary edges [102], [107], [108], [109] with the Hough transform to obtain longer lines. Finally, modeling or template matching [74], [75], [104], [185] is applied to generate the complete railway lines, employing methods such as recursive estimation of spline curves [69], high-order polynomial fitting [121], [124], [125], multiscale modeling [74], [75], [104], and dividing the railway track into subsegments [110] and using different models to approximate different segments.

Traditional image processing methods for obstacle detection can be categorized into moving object detection and static object detection. Common techniques for moving object detection include optical flow [9], [62], [77], frame difference [72], [126], and stereo vision methods [123]. Both the optical flow and frame difference methods identify moving obstacles by analyzing the differences between adjacent images. However, both methods struggle to detect static obstacles due to the minimal difference between the static obstacles and adjacent images. For static object detection, texture feature statistics and template matching methods are used according to the results of rail extraction, which depend on the accuracy of rail detection.

1) *Projected Image-Based Methods:* Karakose et al. [102] installed a camera on a test vehicle to collect images of rail scenes. They applied image preprocessing, Canny edge extraction, and Hough transform detection methods to identify the railroad tracks. They also determined whether the train was running on a straight section or a turnout based on the angle of

TABLE IV
SUMMARY OF PUBLICATIONS EXPLORING TRADITIONAL IMAGE PROCESSING METHODS

Ref.	RD	OD	Rail detection method	Obstacle detection method	Main points	Performance	Computational cost	Suitability
[102]	√	○	Canny edge extraction + Hough transform detection	/	Determine turnouts section	Average success rate: <u>75.3%</u>	CPU, 640×480 pixels	Normal lighting
[103]	√	○	Morphological process (close)	/	Determine the distance and faults	-	CPU, 10 FPS at 1280×720 pixels	Sunny
[104]	√	√	Prior knowledge-based algorithm + modeling	Difference between two adjacent windows	Iterating adaptive bottom-up windows	Average success rate: <u>100%</u>	CPU, 25 FPS at 640×360 pixels	Sunny and shadow
[71]	√	○	HOG feature extraction and region-growing algorithm	/	Determine turnouts section	-	CPU, 10 FPS at 1600×1200 pixels	Cloudy, night with searchlight
[60]	○	√	/	The pixel value difference between the target edge and background	Pixel difference	-	CPU, radar: 187 ms; camera: 33.3 ms	Radar: rain/snow/fog; Vision: clear weather
[74] [75]	√	○	Edge extraction + modeling and pattern matching	/	A semiautomatic method for short distance; an iteration algorithm for long-distance detection	Window error rate: <u>0.42%</u> ; Centerline error rate: <u>3.21%</u> ; Short-distance error rate: <u>0.13%</u>	CPU 8GB, 17 FPS at 768×432 pixels	Sunny and cloudy; but degrades in rainy and rusty rail scenarios
[77]	○	√	/	Optical flow	Moving object detection	Detection distance: $\geq$ <u>600m</u>	CPU	Outdoor lighting fluctuations and motion blur
[107]	√	√	Canny edge extraction (automatically select) + Hough transform detection	Consistency of rail edge gradients		Detection distance: $\geq$ <u>100m</u>	CPU, 25 FPS	Sunlight and tunnel shadows
[108]	√	○	Canny and Sobel edge extraction + Hough transform detection	/		-	CPU	Outdoor lighting variations
[109]	√	○	Sobel edge extraction + Hough transform detection + pattern matching	/	Inverse perspective mapping	Gauge detection error: $\pm$ <u>8 pixels</u>	CPU	Outdoor lighting variations
[110]	√	○	Topology-guided parameter-based methods	/		F1: <u>91.62%</u> at rail-track level, <u>90.73%</u> at all-track pixel level, and <u>96.33%</u> at TP-track pixel level	-	Varying lighting conditions
[111]	√	○	Adaptive IPM transformation algorithm	/		Frame rate: <u>100%</u> in well scenes; <u>43%</u> in abandoned scenes	GPU; 29 FPS	Varying lighting
[112]	√	○	Prior knowledge-	/		Accuracy:	CPU,	Vegetation and

TABLE IV
(Continued.) SUMMARY OF PUBLICATIONS EXPLORING TRADITIONAL IMAGE PROCESSING METHODS

			based algorithm + 2nd order polynomial fitting				97.81%;	1296×966 pixels	snow, but struggle with low light
[69]	√	○	Kalman filter- based recursive estimation	/	Determine turnouts section		Detection rate: 66.7%-92.9% on 4 turnouts	CPU, 10 FPS at 640×480 pixels	Normal lighting, but degree in strong reflections
[70]	√	○	Probabilistic spline curves fitting	/	Determine turnouts section		-	10Hz at 1392×1040 pixels	Normal lighting
[8]	√	○	Regression branch with bottom-up spatial clustering	/	Parameter- based method	Video	On RailSem19 dataset, TP-pixel AP: 92.9% and recall: 88.64%; all-pixel AP: 84.51%, recall: 87.56%	GPU; 12 FPS	Depending on RGB image quality
[9]	○	√	/	Optical flow	Level crossing		Detection rate: 100%; Danger degree error ≤5%	CPU, 5.9GHz WAVE module	Normal lighting
[62]	○	√	/	Texture feature statistic + Optical flow	Video (three CCD cameras, infrared radar)		Detects 0.4 m ² objects up to 250m; Pedestrians in warning vests up to 300m	CPU	Normal lighting; Suburban trains (≤100 km/h)
[114]	○	√	/	Stereo vision and thermal imaging	Stations	Video	-	-	Train-induced illumination; Thermal imaging for biological/non- biological distinction
[115]	√	○	Gabor wavelet- based edge extraction + polynomial fitting	/			Detection rate: 90%; Accuracy: 79.3%; Misdetec- tion rate in mixed snow-shadow scenes: 5%	CPU, 25-50 FPS at 1920×1080 pixels	Rain, snow, rust, curves
[116]	√	○	Polynomial fitting + MLP network	/	Continuous frames		Accuracy: 99.27%	CPU	Normal lighting, sunny, rainy, and cloudy; limited to tunnels or full occlusions
[72]	○	√	/	Frame difference			Detection rate: 80%, up to 232 m	CPU, 1920×1440 pixels	Normal lighting
[117]	○	√	/	Frame difference			Accuracy: 85.8%	CPU, 10 FPS at 1920×1440 pixels	Normal lighting

*Note: RD – Rail Detection; OD – Obstacle Detection.

the two railway tracks at the switching intersection. Based on the above work, Karakose et al. [103] used a morphological process (closing) to enhance rail features and obtain rail contours. They then calculated the pixel value changes in the linear morphology of the rail to determine the distance and detect multiple faults. Wang et al. [104] developed a rail detection function using prior knowledge of rail structure images. They first identified the starting point of the rail and subsequently detected the rail by iterating adaptive bottom-up

windows. When the angle change of the rail line between two adjacent windows exceeds a certain threshold, it indicates the presence of an obstacle. Qi et al. [71] used the HOG feature extraction method to calculate the gradient histogram of the local region (block) of the rail and construct an integral image. Then, they used a region-growing algorithm to extract the rail and determined the train's path based on the open direction of the turnout. Ukai et al. [60], Nassu and Ukai [74], [75], Ukai [77], and Ukai et al. [105] considered the curvilinear

TABLE V
SUMMARY OF PUBLICATIONS EXPLORING DL METHODS

Ref.	RD	OD	Method	Improvements	Performance	Computational cost	Suitability
Two-stage object detection method							
[76]	✓	○	FR-Net with depthwise-pointwise convolution, coarse detection module, and object detection module.	Good tradeoff both on effectiveness and real-time performance	Accuracy: <u>89.53%</u> mAP; 72.3 FPS at 320×320 pixels	GPU; model size 74.2M	Works in rain, fog, and night conditions; stable for railway curves, pedestrians, and trains
[59]	✓	○	FE-SSD inherits a prior detection module of RON, and a feature transfer block of FB-Net employs a novel receptive field-enhancement module	Enhance feature discrimination and robustness; Improve small-scale obstacle detection accuracy	Accuracy: <u>89.5%</u> mAP; 38 FPS at 320×320 pixels	GPU; model size 141M	Works in tunnels, rain, and night; stable for partially occluded and distant objects
[118]	✓	○	Faster R-CNN with ResNet backbone	Improve detection accuracy	Accuracy: <u>89.06%</u> mAP, Waring accuracy: <u>94.85%</u> ; 11.5 FPS	GPU	Works on normal lighting in outdoor railway scenes
[119]	✓	○	Faster R-CNN pre-trained on a public dataset	Improve detection performance	Accuracy: <u>83%</u>	-	Adapts to varying lighting
[11]	✓	✓	Mask-RCNN with ResNet101 backbone	Segmentation	Accuracy: <u>95.7%</u> mAP; 5.56FPS at 1920×1080 pixels	GPU	Work in subway tunnels
[68]	✓	○	Mask R-CNN with FPN neck	Improve detection accuracy; Segmentation	On the RSDS dataset, Pixel Accuracy: <u>91.6%</u> ; MIOU: <u>89.8%</u> ; F-measure: <u>86.8%</u> ; 20FPS at 1920×1080 pixels	GPU	Adapts to varying lighting and complex backgrounds; low-speed autonomous trains (≤60 km/h)
[121]	✓	✓	Mask-RCNN with SSwin-Le Transformer backbone and ME-PAFPN neck	Improve multi-scale obstacle detection performance; Segmentation	Accuracy: <u>91.3%</u> mAP; 4.2FPS	GPU/CPU; 59.3M parameters and 322.01 GFLOPs	Work in normal weather (sunny, rainy, night)
[122]	○	✓	Key frame extraction; a two-stage Transformer network with an image masking strategy and dual-channel network	Reconstructing differences to locate anomalies; Enhance multiclassification accuracy	On IRSA dataset, precision: <u>97.8%</u> , recall: <u>96.6%</u> , AUC: <u>99.3%</u> ; On RTAD dataset, precision: <u>96.7%</u> , recall: <u>95.4%</u> ; 62.5 FPS at 224×224 pixels	GPU/CPU; 7.92M parameters and 322 GFLOPs	Work in normal lighting
One-stage object detection method							
[123]	✓	✓	Improved SSD with ResNet backbone	Improve small object detection performance	Accuracy: <u>91.61%</u> mAP; 26.3 FPS at 1280×720 pixels	GPU/CPU	Work in diverse lighting (day/night) and weather (sunny/rainy)
[124]	✓	✓	Improved YOLOv4 with D-CSPDarknet backbone and SPP neck	Improve detection accuracy and efficiency	Accuracy: <u>93%</u> mAP; 7.19 FPS at 416×416 pixels	GPU/CPU	Work in normal lighting
[6]	✓	○	FE-YOLO with an attention module and SPP and BCPANet neck	Improve real-time performance	Accuracy: <u>92.57%</u> mAP; 10.1 FPS at 416×416 pixels	GPU; 201.63M parameters	Work in fog, rain, and strong light
[125]	○	✓	FFE-Net with anchor frame filtering module (AFM), feature fusion enhancement module (FFE), and refinement detection module (RDM)	Improve small-scale obstacle detection accuracy	Accuracy: 88.55% mAP; 26 FPS at 320×320 pixels	GPU	Work in normal weather (sunny, rainy, night)
[64]	✓	✓	DFF-Net based on SSD, and prior module	Improve multi-scale obstacle detection accuracy	Accuracy: <u>90.12%</u> mAP; 54 FPS at 320×320 input	GPU; Model size 109.5M parameters	Work in tunnels, rainy days, and strong light
[126]	○	✓	YOLOv5 with convolutional block attention module (CBAM)	Improve small and occluded obstacle detection accuracy	Accuracy: <u>94.1%</u> mAP; 39 FPS	GPU	Work in insufficient lighting or adverse

TABLE V
(Continued.) SUMMARY OF PUBLICATIONS EXPLORING DL METHODS

			and CIOU loss				weather
[127]	○	✓	YOLOX-tiny model with coordinate attention (CA), CIOU, and focal loss; RGB-d-OCR Matching algorithm for distance measurement	Improve detection accuracy; Measure the distance	Accuracy: <u>55.34%</u> mAP; Equipment recognition rate: <u>91.88%</u> ; ID recognition rate: <u>88%</u> ; Binocular ranging error <u>≤2%</u> ; 15 FPS	GPU; 5.05M parameters with low memory footprint	Work in normal lighting and weather
[128]	✓	✓	Yolov8n model with dynamic sparse attention module (D-BRA); DeepSORT with Swin Transformer for trajectory-based false detection filtering	Improve small and occluded obstacle detection accuracy	Accuracy: <u>86.57%</u> mAP@0.5; False detection rate reduced by 16% with DeepSORT; 105.5 FPS	GPU; 6.2M parameters and 32.3 GFLOPs	Work in occlusion, rain/snow, and lighting variations
[129]	○	✓	Swin Transformer-based dual-branch parallel fusion network	Enhance recognition of targets of varying sizes	On FIRA dataset, MIOU: <u>81.34%</u> , F1 score: <u>78.95%</u> ; On LandCover dataset, MIOU: <u>78.57%</u> , F1 score <u>74.61%</u> ; 79 FPS	GPU; 76.79 M parameters and 61.05 G FLOPs	Work in complex backgrounds
[130]	○	✓	DETR framework with RetinexNet image enhancement network and Global Channel Attention (GCA) module	Enhance low-light images	Accuracy: <u>77.34%</u> mAP; 32.43 FPS	GPU	Work in normal lighting and dark time
[131]	○	✓	RT-DETR-r18 with Lightweight Block EMA, and fuse Context Anchor Attention and Dynamic sampling module	enhances small-target detection speed and accuracy	Accuracy: <u>86.5%</u> mAP@50; 140.8 FPS	GPU, 19.06M parameters, 62.2 GFLOPs	Work in normal lighting
[132]	○	✓	Improved yolov5 with ShuffleNetv2 or GhsotNet backbone	Lightweight	On COCO dataset, accuracy: <u>90.3%</u> mAP@50; On custom railway dataset, accuracy: <u>88.2%</u> mAP@50; 109-111 FPS	CPU; Model size 2.2MB; 940K parameters and 1.9 GFLOPs	Work in day/night, rain/fog scenarios, integrating LiDAR, telephoto/close-focus camera
[133]	✓	✓	Improved yolov4-tiny with CSP-ShuffleNet backbone, MHAM in neck and depthwise separable convolution in head	Lightweight	On COCO dataset, accuracy: <u>41.5%</u> mAP; On custom railway dataset, accuracy: <u>66.3%</u> ; 17.7FPS at 416×416 input	Jetson TX2; Model size 11.4MB; 2.89M parameters; 4.2 BFLOPs	Work in lighting changes (day/night, rain/fog)
[134]	✓	✓	LGF-Net with CSP-GhostNet backbone, attention mechanism	Improve small-scale obstacle detection accuracy	Accuracy: <u>94.93%</u> mAP; 38 FPS	Jetson Xavier NX, Model size 11.4MB; 12.7 GFLOPs	Work in adverse weather (rain/fog/low-light) with starlight camera for weak-light imaging
[135]	○	✓	Improved YOLOv4-tiny backbone with CRP module, IoM metric, and postprocessing	Reduce parameters; Improve inference speed	Accuracy: <u>80.3%</u> mAP; 15 FPS at 800×800 pixels	GPU/CPU; 2.91M parameters and 2.43G FLOP	Work in sunny/dusk lighting; supporting dynamic scenes at inspection speeds of 5-25 km/h
[136]	○	✓	AE-Net with enhanced transformer module, lightweight integration module, and reinforced multiscale fusion module	Improve small and multitarget detection accuracy and speed	Accuracy: <u>95.29%</u> mAP; 145 FPS on CPU	GPU/CPU; 15.3 GFLOPs	Work in fog, night, tunnels, and varying lighting; near-infrared laser lighting for low-light environments
[137]	○	✓	Visual-inertial SLAM with a lightweight deep	Deep feature-based object tracking	ATE of 0.2960 m in real dynamic scenes;	Jetson Orin; 121K	Suitable for fast-moving dynamic

TABLE V
(Continued.) SUMMARY OF PUBLICATIONS EXPLORING DL METHODS

			feature extraction network; object tracking via feature point pair clustering		15-20 FPS	parameters; 178M FLOPs	objects, weak-texture scenes
Segmentation-based method							
[145]	✓	○	U-Net and ResUNet++ with BCE loss	Improve rail detection accuracy	On RailSem19 dataset, U-Net: mIoU: <u>81%</u> , Rail IoU: <u>63%</u> , and Dice score: <u>77%</u> ; ResUNet++: mIoU: <u>80%</u>	GPU, U-Net: 30M parameters; ResUNet++: 14 M parameters	Work in rain, snow, fog, and varying lighting (day/night)
[147]	✓	✓	Improved U-Net integrates ResNet34, MobileNetV2 and VGG16 backbone with a hybrid loss function	Improve rail detection accuracy	On RailSem19 dataset, accuracy: <u>85%</u> mAP; mIoU: <u>54.3%</u>	CPU/GPU	Work in diverse conditions (complex scenes, difficult weather)
[148] [149]	○	✓	U-Net structured autoencoder for obstacle detection	Classify obstacles	On Vesuvio Dataset, anomaly detection module: accuracy: <u>96.6%</u> and F1-score: <u>97.3%</u> ; Localization module: accuracy 90.3% with IoU>0.3 and 7.3 pixels average center distance; Classification module: average F1-score: <u>78.5%</u> ; 190 FPS (detection), 130 FPS (localization), >1000 FPS (classification)	Deployable on edge devices; 28.1M parameters	Work in low-light conditions (night, tunnels) using thermal imaging
[150]	✓	✓	U-Net for rail segmentation; Autoencoder for reconstruction	real-time anomaly detection with low latency; reduces need for annotated anomalies	On RailSem19 dataset, U-Net with VGG mIoU: 52.78%; Reconstruction error detection: 100% true positive/negative rates; 27 FPS	Lightweight autoencoder with 128-dimensional latent space	Work in lighting variations
[151]	✓	○	SegNet with atrous convolution and dilated-cascade connection and cascade sampling	Improve rail segmentation accuracy; lightweight	mIoU: 98.46% and mAP: 99.15%; 38 FPS on NVIDIA Titan X GPU; 3.9 FPS on TX2 embedded platform	GPU/embedded platform	Work in tunnels, elevated areas, and day/night lighting
[152]	✓	○	U-Net with atrous convolution and a channel attention module	Improve rail segmentation accuracy; lightweight	On RAWRail dataset, Accuracy (ACC): <u>98.6%</u> , False Positive Rate (FPR): <u>1.1%</u> , False Negative Rate (FNR): <u>0.7%</u> ; 74 FPS at 640×360 pixels	GPU; Reduce computational load by 2/3 via DWConv	Work in good daytime lighting
[153]	✓	○	U-Net with SPP neck and ARM module	Improve rail segmentation accuracy	Accuracy: <u>94.4%</u> ; IoU (all tracks), <u>91.2%</u> wIoU; 20 FPS	GPU	Work in varying lighting (sunny, overcast)
[154]	✓	○	DFA-UNet with atrous convolution and maximum pooling channel attention module	Improve rail segmentation accuracy	On RailSem19 dataset, mIoU: <u>86.62%</u> , F1-score: <u>89.97%</u> , and Accuracy: <u>99.24%</u> ; 49.61 FPS	GPU; 8.75M parameters and 24.96G FLOP	Work in low-light conditions (night, tunnels)
[157]	✓	○	RailNet with VGG backbone and orthogonal attention mechanism	Improve rail segmentation accuracy	On RailSem19 dataset, achieves mRecall: <u>92%/89%</u> (train/val) and mIoU: <u>59%/54%</u> ; RailNet-LW 5.81 FPS and	At 320×640 input, RailNet-AW occupies 209MB	Work in varying lighting, vegetation coverage, and rail crossings

TABLE V
(Continued.) SUMMARY OF PUBLICATIONS EXPLORING DL METHODS

					RailNet-AW 4.78 FPS on 320×640 pixels		
[158]	✓	✓	DMFNet with dense connections, multiscale fusion, and loss function weight reassigning.	Lightweight; Improve semantic segmentation accuracy	On Railsapes dataset, mIoU: <u>75.11%</u> and 108 FPS; On MRSI Dataset: mIoU: <u>90.34%</u> and 105.51 FPS	3.34M parameters and 27.22 GFLOPs; 108 FPS on RTX 3090 and 13.34 FPS on Jetson Xavier NX (1024×1024)	Work in varying lighting
AE and GAN-based -based methods							
[148]	○	✓	Autoencoder for obstacle detection		Accuracy: <u>96.6%</u> and F1-score: <u>97.3%</u> with thermal images; 190 FPS	GPU; Autoencoder with 22M parameters, classifier with 700k parameters	Rely on thermal imaging for operation in light-free conditions
[150]	✓	✓	U-Net for rail segmentation and Autoencoder for obstacle detection		On RailSem19 dataset, U-Net with VGG mIoU: <u>52.78%</u> ; Reconstruction error detection: <u>100%</u> true positive rate, no false positives/negatives; 27 fps	Lightweight autoencoder with 128-dimensional latent space	Work in lighting variations
[159]	✓	✓	Autoencoder convolution network reconstruction	Compare image differences	Average gap-score <u>68.17%</u> , with peak gap-scores up to 230% in real-world obstacle frames	lightweight design	Rely on RGB images
[161]	○	✓	Improved conditional DCGAN (C-DCGAN) to build obstacle dataset + YOLOv3		Accuracy: Pixel-wise accuracy <u>80.46%</u> , mIoU: <u>65%</u> , mAP <u>69%</u> ; mIoU of real-generated pedestrian pairs: <u>85%</u> ; FID: <u>26.87</u>	GPU; Generation: 327 ms per sample	Suitable for generating pedestrians, livestock; works in various railway scenes via gauge-based scale estimation
[162]	○	✓	Diffusion model		Image-level AUROC: <u>99.05%</u> , pixel-level AUROC: <u>97.62%</u>	GPU	Work in bright, dim, and flickering light sources
[163]	○	✓	GANomaly generative adversarial network		Pixel-level AUROC: <u>97.62%</u> , image-level AUROC: <u>99.05%</u> ; Same-line detection: <u>90%</u> recall	GPU	Certain robustness to lighting changes
[164]	○	✓	Improved SRGAN generative adversarial network + SSD	Increase image quality; Improve obstacle detection accuracy	On Urban100 dataset with 4× upscaling, PSNR of <u>26.01</u> and SSIM of <u>0.7637</u> ; OPENPOSE operates at 8.6 FPS	GPU	Robustness to low-light conditions
Few-shot based methods							
[165]	✓	✓	Few-shot learning with shuffling labels and neuron dropout.	Improve detection accuracy to new environment	Accuracy <u>83.8%</u> with 5-shot (K=5), FPR <u>6.8%</u> , and FNR <u>9.4%</u> ; Maintains <u>79.46%</u> accuracy in low-light conditions; 29 FPS on 640×480 input	30,434 parameters, 24.96G FLOPs	Work in varying lighting (day/night)
[166]	✓	✓	few-shot object detection with a contractive head and a classifier head initializer	Improve detection accuracy, adaptability, and generalization to new environment	Accuracy: <u>97.96%</u> , precision: <u>98.61%</u> , recall: <u>98.07%</u> , and F1-score: <u>98.34%</u>	Computational efficiency is optimized via local feature	Work in lighting changes

TABLE V
(Continued.) SUMMARY OF PUBLICATIONS EXPLORING DL METHODS

					concatenation and PSA module		
Track trajectory detection methods							
[90]	✓	○	A row-based rail detection framework containing a lightweight convolutional backbone and an anchor classifier	Row-anchor-based methods	On Rail-DB dataset, accuracy: <u>92.77%</u> , with <u>96.22%</u> in night scenes and <u>95.82%</u> in straight lines; 312 FPS, 1.64x faster than segmentation methods	8.42G FLOPs and 182.84MB memory usage	Work in diverse environments
[168]	✓	○	ResNet18/34/50 with two specialized branches: Horizontal Coordinate Locator and Perspective Identifier		On SRail dataset, <u>98.68%</u> F1@50; On RailDB dataset, <u>97.43%</u> F1@50 and <u>67.70%</u> F1@75	ResNet18 backbone has 11.7M parameters; 473 FPS with ResNet18	Work in complex environments (tunnels, shadows)
[169]	✓	○	Projection length-based rail representation with inter-rail attention		mF1: <u>73.68%</u> , F1@50: <u>79.25%</u> , and FPS >38	ResNet18 with 11.7M parameters, 3.2 GFLOPs	Work in adverse weather (rain, snow, fog) and complex backgrounds (trees, buildings)
[171]	✓	○	Line-based methods with an attention mechanism and a semantic segmentation head	Line-anchor-based methods	On RailwayLine dataset, achieves <u>94.94%</u> accuracy and <u>94.77%</u> F1-score, false positive rate <u>2.55%</u> , false negative rate <u>8.21%</u> ; 58 FPS on 800×320 pixels	Moderate parameters: 22.29M and 20.92 GFLOPs.	Suitable for urban rail transit and railway inspection, supporting straight/curved rail detection in complex environments with occlusions and lighting variations
[172]	✓	○	Dynamic anchor line-based rail detection with ResNet backbone, dynamic anchor line generator, and pyramid pooling module		On DL-Rail dataset: <u>96.43%</u> F1@50, <u>59.79%</u> mF1, 93.23% AP@50, <u>3.96px</u> X-error@50, and 212 FPS; <u>93.92%</u> F1@50 in rainy conditions, <u>94.33%</u> in nighttime, and <u>88.33%</u> in turnout areas	11.7M parameters and 9.75 GFLOPs with ResNet18 backbone	Work in rain, snow, and nighttime conditions, with strong performance under complex backgrounds (vehicle/pedestrian occlusion)
Camera-based distance estimation method							
[173]	✓	✓	Monocular; Train location estimation using multidimensional geometric constraints	Distance estimation	Achieves <u>95.5%</u> , RMSE on the L-Short sequence: <u>37.07m</u> , RPE MEAN on H-Median: <u>5.12m</u> ; Speed estimation error <u>≤3.04%</u> in low/medium-speed scenes (L-Short/L-Median), <u>0.62%</u> in high-speed (H-Median)	Per-frame processing time 72.91-83.46 ms (track detection 22.91-23.89 ms, object detection 12.05-12.59 ms), frame rate >10 Hz; memory usage ≤200 MB	Work in GNSS-denied environments (tunnels, viaducts); Reduced object matching success at high speeds (>17 m/s)
[78], [79]	○	✓	Multi-DisNet distance estimator + YOLO object classifier		Human target detection errors: <u>5.31%</u> - <u>13.22%</u> ; Reliable distance: <u>600m</u>	GPU	Work in diverse lighting (day/night, dusk)
[81]	○	✓	Multi-camera distance estimation system based on a multilayer neural		Long-Range Accuracy: MAE <u>22.76m</u> and RMSE	Processes synchronized RGB/thermal	Work in diverse weather (day/night, temperature)

TABLE V
(Continued.) SUMMARY OF PUBLICATIONS EXPLORING DL METHODS

			network		<u>34.09m</u> for 0-1000 m distances; Estimate distances within 5 m error on moving trains Foreign object intrusion warning accuracy: <u>95.65%</u> , distance prediction accuracy: <u>96.22%</u> , and area estimation accuracy: <u>90.72%</u>	frames at 2 fps	variations)
[174]	✓	✓	Self-attention distillation SAD algorithm + YOLOv4			GPU	Work in all-weather detection; Millimeter wave radar has strong penetration through fog, smoke, and dust Suit 500-2000m braking distances; Work in complete darkness (0 lux) and track inclinations; Thermal imaging is insensitive to rain/snow/fog
[175]	✓	✓	Integrates image-plane homography with fuzzy logic to optimize obstacle distance estimation using thermal imaging		Max error: <u>7.1%</u> in 400-950m range; <u>0.82%</u> error at 850m	60 FPS operation with 3.65×10^{-6} s average execution time	
LiDAR and radar-based method							
[176]	✓	✓	Image template matching method + laser measurement	Turnouts detection	Achieves <u>92.5%</u> correct obstacle detection in simulations; Laser range finder operates within 50-70 m	Low-complexity mask matching algorithm runs on embedded systems	insufficient adaptability to severe weather (rain, fog)
[177]	✓	○	FarNet for long-range rail track point cloud segmentation; Converts LiDAR point clouds into pseudo range images to enable 2D CNN processing	Long-range detection	MIoU: <u>98.0%</u> ; MPA: <u>98.9%</u> on rail track segmentation	-	Work in straight, curved, uphill/downhill tracks, but struggles with turnouts; Works in varying lighting (tunnel/elevated) Suitable for a 300m range distance and stable in tunnels/viaducts; rely on uniform LiDAR point distribution Suitable for switchbacks, tunnels, curves, straightaways, robust to bad weather and complex lighting
[178]	✓	✓	A stacked-hourglass network backbone with a global-local feature fusion module and a pooling-guidance mechanism	Enhance depth estimation from sparse LiDAR data	RMSE: <u>8165.673mm</u> and MAE: <u>2596.418mm</u> on the Hong Kong Tsuen Wan Line dataset	313.5 Mb model size with 27.74 ms/frame inference time	
[179]	✓	✓	A multimodal OID method based on image semantic segmentation + rail track topology reconstruction + LiDAR point cloud analysis	Improve track reconstruction accuracy through post-processing	Overall Recall: <u>98.2%</u> , F1: <u>90.9%</u> , Precision: <u>90%</u> , Accurate: <u>99.9%</u>	0.27s/frame processing on Jetson AGX Xavier, CPU utilization 80%	
[180]	✓	✓	A lightweight camera and lidar fusion framework: a D-YOLO for 2D obstacle detection + coordinate transformation extracts lidar point cloud features for distance estimation	Enhances feature representation through dual-feature fusion	Overall mAP: <u>85.1%</u> ; Detection speed: <u>94 FPS</u>		Work in both daytime and nighttime conditions, real-time detection
[5]	○	✓	Camera and Lidar localize ROI area	Long-range detection	RMSE at 46 m in indoor tests: <u>6.179</u> mdeg; Precise targeting for 1D LiDAR at >1,000 m	Low-cost hardware (Arduino, ROS)	Suitable for long-range obstacle detection in tunnels and viaducts; Stable in controlled environments Suitable for train frontal obstacle detection, effective in daytime and
[181]	✓	✓	Use camera alone in daytime; Fuse LiDAR point cloud to supplement object-likeness in low-	Reduces over-detection to <2% by limiting detection area centered on rail vanishing point	Daytime: >98% detection rate in 300 m for persons; Low-light: >90% detection		

TABLE V
(Continued.) SUMMARY OF PUBLICATIONS EXPLORING DL METHODS

			light/nighttime; Limit detection area via rail extraction algorithm		rate at 200 m; Nighttime (0.2-0.3 lx): >90% within 150 m, 45% at 200 m MobileNetV2 achieves <u>99.55%</u> accuracy on Event Volumes, with precision <u>99.49%</u> and recall <u>99.65%</u> ; False alarm rate $\leq 1\%$; near 0 miss rate in scenarios with train lights, rain/snow, and animal intrusions		nighttime scenarios
[94]	○	√	Dynamic Vision Sensor architecture with event-volume encoding and spatiotemporal filtering mechanism		Event encoding at 20 FPS		Work in extreme conditions (strong/zero light, rain/snow) at tunnel entrances; resists interference from train light reflections and insect movements
[182]	○	√	Optical flow network uses U-Net structure; TTI network fuses flow and depth via residual branches, with the dynamic version incorporating temporal depth frames for enhanced motion estimation	Event camera	Optical Flow: 0.36 / 15.69 (EPE, event/non-event); TTI: 0.94 / 0.89 (P/R, static); Evasion: 2.65° / 4.93° (angle, normal/high-risk)	-	Suitable for dynamic obstacles and low-texture environments; Integrates event camera's high dynamic range with LiDAR depth information
[183]	○	√	Iterative event-based Hough Transform with adaptive NMS; spatio-temporal HT for cross-frame tracking; odometry-integrated triangulation for 3D mapping		Average event processing time <u>0.65</u> μ s/event, real-time factors <u>0.41-0.59</u> , supports 204,909 events/s data streams.	-	High-speed trains (≤ 25 m range), Work in dynamic lighting (day/night/tunnels), close-range vertical structure (power poles, signal poles) detection

*Note: RD – Rail Detection; OD – Obstacle Detection.

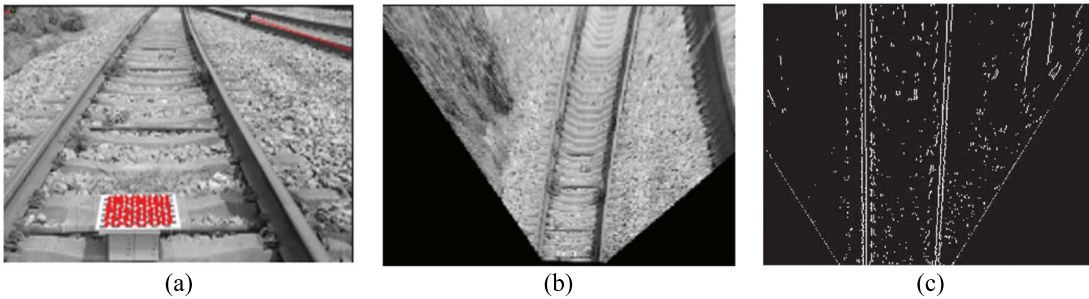


Fig. 5. BEV-based methods. (a) Camera intrinsic calibration. (b) IPM image. (c) Edge map with the modified Sobel operator [108].

changes in the rail morphology at a distance and matched the rail extraction results with standard rail patterns or models. They achieved accurate rail detection by extracting feature points using SIFT operation or by modeling each rail as a sequence of parabola segments: a single segment for short distances and multiple segments for long distances. Obstacles were detected based on differences between the object pixel edge values and the background pixel values, while the optical flow method was used to detect moving objects. In addition, the fusion of image sensors and radar was used to coarsely locate the position of obstacles, thereby improving real-time detection performance and accuracy.

2) *BEV-Based Methods*: Unlike the perspective view captured by the aforementioned onboard cameras, which deform objects based on their position, the bird's-eye orthogonal view (inverse perspective mapping (IPM) [106]) maintains a constant width of the trajectory throughout the image. This allows for a more direct assessment of all the geometric constraints required by the algorithm, as illustrated in Fig. 5. To fully utilize the parallel and equidistant characteristics of the rail, Maire and Bigdeli [107] and Wang et al. [108] employed inverse projection transformation to convert images captured by onboard cameras into a BEV. This transformation preserves the parallel and equidistant nature of the paired

rails. Subsequently, rail line detection was performed using the Hough transform based on rail edge features, and obstacle intrusion was determined by the continuity of the rail line. The main difference between these two approaches is whether the camera is calibrated or not. Based on the previous work, Wang et al. [109] utilized a pattern-matching method to extract rail tracks by matching the edge features of the actual image with parametrically modeled candidate rail patterns. Yang et al. [110] proposed a topology-guided method that segments rail pixels, applies IPM for pixel matching, and reconstructs rail lines using geometric constraints, achieving an $F1$ score of 91.62% on the Hong Kong Metro Tsuen Wan Line. To address the low generalization of conventional IPM due to fixed camera parameters, Lin and Peng [111] introduced an adaptive IPM approach that dynamically adjusts perspective parameters by detecting trapezoidal calibration regions formed by rails and sleepers. Gschwandtner et al. [112] adopted the inverse projection transformation to select the correct candidates by recursively combining rail candidates from different slices and applied a second-order polynomial fitting for rail detection. A limitation of the IPM method is that it requires accurate camera calibration, which cannot be guaranteed in practice applications, making it unsuitable for long-distance railway detection.

3) *Consecutive Frame-Based Methods*: In addition to single-frame image analysis, Ross [69] proposed constructing a discrete Kalman filter model for rail geometry estimation, emphasizing the continuity of rail detection across video frames. They used recursive estimation to track the rail trajectory in images based on the detected rail edges in different frames. Further to the above work, Ross [70] represented the rail as probabilistic spline curves and used a recursive estimation algorithm to iteratively optimize the rail in each frame to determine whether it is in a straight-line or turnout section. Ghorbanalivakili et al. [8] proposed TPE-Net, a consecutive frame-based rail detection method that regresses rail centers and pixel probabilities using a two-branch network and reconstructs continuous trajectories via clustering, polynomial fitting, and topological filtering. Salmane et al. [9] proposed a video analysis-based method for obstacle detection at LCs. This method first extracts the obstacle area through optical flow segmentation and detects moving objects. Then, based on the center of the bounding box corresponding to the object, a hidden Markov model is designed to estimate the ideal trajectory of the object center. Ruder et al. [62] proposed a method to detect obstacles in front of the train by using image edge and texture features combined with the optical flow method and then applied the Kalman filter [113] to track the obstacles within the images. Oh et al. [114] proposed a passenger safety monitoring system for railway stations, which includes an information fusion module that analyzes video input from cameras and uses a stereo vision algorithm to locate obstacles within the railway area. In addition, Selver et al. [115] divided a video frame into four subregions from far to near and used multiscale and multidirection Gabor wavelets to extract the rail edges of each region. They employed linear and second-order polynomial fitting to model the rail profile. Furthermore, Selver et al.

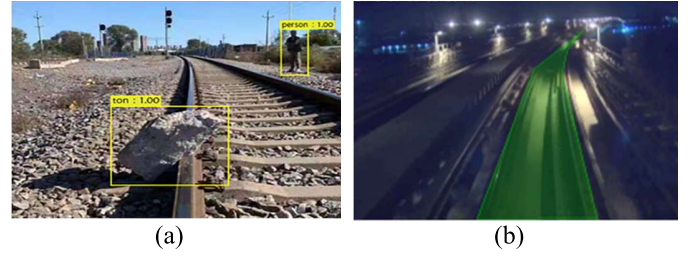


Fig. 6. DL-based methods. (a) Obstacle detection [126]. (b) Rail track segmentation [151].

[116] proposed a rail extraction method based on statistical properties. First, a polynomial fitting procedure is applied to analytically represent various railway lines, and then, a multilayer perceptron (MLP) network is introduced to predict railway lines through polynomial least-squares approximation and regression based on statistical analysis coefficients. Based on the frame difference method, Nakasone et al. [72] proposed real-time communication between the front and rear trains to compare images captured by the front train when passing the same section with the currently captured images for obstacle detection. Mukojima et al. [117] computed frame-by-frame correspondences between the current image and a database of prephotographed images along the rail, and applied image subtraction to these corresponding frames to detect obstacles.

B. DL-Based Methods for OID

The methods discussed above primarily rely on handcrafted features or traditional machine learning techniques. However, their performance depends heavily on the choice of threshold and seed pixel values, making them susceptible to issues such as varying illumination, noise, and occlusion. Consequently, the applicability and robustness of these methods may not be sufficient to address the challenges posed by complex and diverse railway scenarios. With the rapid development of AI technology, numerous high-performance DL neural networks have been applied to various visual inspection tasks. The railroad tracks have high-level semantics and local features of slender geometry, while obstacles vary in size. Therefore, effective OID methods must exploit both low- and high-level features, as well as local and global context, to accurately identify railway tracks and obstacles. Deep neural networks excel at extracting multilevel features. The low-level layers extract specific high-resolution features, such as contours and scales, which are beneficial for object localization. The high-level layers, on the other hand, extract abstract low-resolution features rich in semantic information, which are crucial for classification. Several DL-based object detection and segmentation frameworks leveraging these features have been proposed for rail OID with remarkable results. Fig. 6 illustrates the representative results: obstacle detection and rail-track segmentation. Researchers are working on refining the structure of each subnetwork within these frameworks to improve performance, and strategies such as multiscale feature fusion, attention mechanisms, lightweight network design, and specialized loss functions have been adopted. In addition to the

classical frameworks, emerging DL architectures such as generative adversarial networks (GANs) and autoencoders (AEs) have also been applied to obstacle detection. Furthermore, DL-based few-shot learning methods have been introduced to address the issue of limited data in real-world OID applications.

1) *DL-Based Object Detection Frameworks:* Object detection algorithms identify and localize objects by generating bounding boxes in images or video sequences to indicate the location of the objects. Both two- and one-stage object detection frameworks, as well as Transformer-based detection frameworks (e.g., DETR and its variants), have been used for OID in railways.

Based on the two-stage object detection framework, Ye et al. [59], [63] proposed a CNN-based rail detection approach by classifying rail regions into three categories: straight rail, left-turn rail, and right-turn rail. The backbone network employs depthwise-pointwise convolution to accelerate detection. Coarse detection is performed using the top layer features to regress anchor positions and sizes, which are then exploited in the head network to improve detection accuracy. Yu et al. [118] utilized ResNet to build a backbone network based on Faster R-CNN, which improved the accuracy of obstacle detection. Kapoor et al. [119] transferred a Faster R-CNN model pretrained on a public dataset and fine-tuned it on rail images captured by an IR camera to enhance detection performance. Mask R-CNN builds on the Faster R-CNN framework by adding a branch for generating pixel-level masks of objects for instance segmentation. He et al. [11] extended the Faster R-CNN using the Mask R-CNN [120] network with ResNet101 as the backbone network to simultaneously detect and segment both rails and obstacles. To improve the accuracy of rail segmentation, Wang et al. [68] used FPNs to fuse features of different scales based on the Mask R-CNN network. He et al. [121] proposed a novel network ME Mask R-CNN based on the Mask R-CNN framework. They introduced the spatial shuffle and depth-separable convolution techniques into the Swin Transformer to construct the backbone network. Moreover, they used the path aggregation network ME-PAFPN for feature fusion and integrated multiscale enhancement methods to improve the detection accuracy of small objects. Li et al. [122] proposed a two-stage Transformer network, AnoDet, for anomaly detection in intercity rail transit. Their approach utilizes key frame extraction and image masking methods to perform image restoration on abnormal samples and locate abnormal areas by reconstructing difference scores. The located abnormal areas are combined with other abnormal images to form an anomaly set using knowledge migration and input into the classifier to assign abnormal category labels.

Within the one-stage object detection framework, Xu et al. [123] proposed an improved SSD rail obstacle detection network. They first introduced ResNet into the backbone network and then increased the network depth to enhance the extraction performance of top-level semantic features. Furthermore, they exploited multiscale features in the output layer for object detection. He et al. [124] proposed an improved YOLOv4 rail obstacle detection network. They designed a new feature extraction network D-CSPDarknet, which combines

path aggregation and FPNs for effective feature fusion. They also implemented a spatial pyramid pooling (SPP) network in each fusion layer, solving the problems of low accuracy and poor real-time performance of traditional detection methods. He et al. [6] proposed a flexible and efficient multiscale one-stage object network FE-YOLO for obstacle detection. The feature extraction network consists of an attention module, a downsampling module, a residual module, and an SPP module. A repeatable bidirectional cross-scale path aggregation module is employed for feature fusion. Yang [125] proposed a feature fusion enhancement network FFE-Net to improve the accuracy of small-scale obstacle detection. The network consists of three parts: anchor frame filtering module (AFM), feature fusion enhancement module (FFE), and refinement detection module (RDM). The FFE enhances the detection performance of small and large objects in vertical and horizontal comparisons. The AFM and RDM address the imbalance between positive and negative samples during network training through cascade regression. Ye et al. [64] proposed a DFF-Net rail segmentation network, which incorporates different fusion structures across four output layers to improve the detection accuracy of objects of various scales. Wang and Yu [126] proposed adding a CBAM to YOLOv5, allowing detected objects to receive greater attention in both channel and spatial dimensions while minimizing the influence of weakly correlated features. This method enhances object representation and reduces missed detections of small and occluded objects. Furthermore, they replaced the GIOU loss in the location regression loss function with the CIOU loss to improve the convergence speed of the model. Liu et al. [127] proposed a CNN and binocular vision-based framework for detecting trackside equipment in railway systems. They enhanced the YOLOX-tiny model with coordinate attention (CA), CIOU, and focal loss, and integrated results with binocular ranging and OCR recognition using the RGB-d-OCR Matching algorithm. The system is deployed on a track inspection vehicle to automatically position and measure the distance to determine whether the equipment violates boundary constraints. Wang et al. [128] proposed BCD-YOLO, an improved YOLOv8-based method for railway perimeter foreign object intrusion detection. The approach integrates an enhanced dynamic bi-level routing attention (D-BRA) module for fine-grained feature extraction, a T-CARAFE operator to reduce upsampling feature loss, and an improved DeepSORT with Swin Transformer for trajectory-based tracking to filter false alarms. Experiments show that BCD-YOLO achieves 5.22% higher precision and 2.35% better mAP@0.5 than YOLOv8n, with a 16% reduction in false detection rate when combined with DeepSORT.

Based on Transformer-based detection frameworks, Ye et al. [129] proposed DPFNet, a dual-branch parallel fusion network using Swin Transformer for global context and a dual-branch structure to process image patches of different scales, to classify and extract railway hidden dangers via semantic segmentation, outperforming classical models on FIRA and LandCover datasets. Zheng et al. [130] proposed Re-DETR for fast detection of railway engineering targets in dark time domains, combining RetinexNet for image enhancement (decomposing illumination and reflection components, and

adjusting brightness/contrast) with an improved DETR model that integrates a global channel attention (GCA) module to suppress noise and a CIoU-based loss function for accurate bounding box regression. Liu et al. [131] evaluated RT-DETR, YOLOv5, and YOLOv8 on a self-built railway perimeter intrusion dataset, training YOLOv5 and YOLOv8 for 300 epochs and RT-DETR for 72 epochs. By comparing their performance in terms of accuracy and FPS, the study verifies that RT-DETR is suitable for railway perimeter intrusion detection, especially for long-distance small target recognition.

While earlier approaches often relied on conventional backbones that compromised speed and resource efficiency, the growing demand for faster and more efficient models has shifted attention toward lightweight architectures. In view of this, lightweight backbones are now increasingly considered to balance performance with efficiency. Chen [132] proposed a lightweight network based on YOLOv5, where they used two lightweight structures, ShuffleNetv2 and GhostNet, to replace the backbone network of YOLOv5. Guan et al. [133] proposed an improved YOLOv4-tiny lightweight network by constructing a CSP-ShuffleNet lightweight structure and integrating the cross-stage structure and channel blending strategy into the backbone feature extraction network to accelerate inference speed. In the top feature layer of the backbone network, they implemented a multihead attention mechanism to enhance object localization capabilities. Moreover, in the head network, traditional convolutions were replaced by depth-separable convolutions, further reducing network parameters and deployment costs. Zhao et al. [134] proposed an enhanced Transformer structure in the backbone, incorporating CSP-GhostNet into the Transformer to reduce the computational cost and improve the feature extraction capability of the model. This combination leads to higher detection accuracy compared to YOLOv5. Guan et al. [135] proposed a lightweight three-stage framework for railway obstacle detection, combining a coarse RP (CRP) module, an improved YOLOv4-tiny network (RODNet), and postprocessing. RODNet reduces parameters by 48% and boosts the inference speed by 68.2%, offering competitive performance against larger models. Zhao et al. [136] proposed AE-Net, a convolution- and Transformer-based network for real-time railway obstacle detection. It features three modules: enhanced and lightweight Transformer module (ETM) for global contextual modeling, lightweight feature integration module (LIM) for efficient feature integration, and reinforced multiscale feature fusion module (RFM) for better small obstacle detection. AE-Net outperformed YOLOv5s in both accuracy and speed, demonstrating robustness for multitarget detection in complex railway environments. Zhang et al. [137] proposed a real-time dynamic visual-inertial SLAM and object tracking system using a lightweight deep feature extraction network to improve feature detection. The system tracks multiple dynamic objects without semantic data by clustering feature point pairs and includes an incremental loop closure detector to enhance global pose optimization, improving accuracy and object tracking performance.

2) *DL-Based Segmentation Frameworks*: Segmentation-based methods convert the labeled polylines into multivalued masks and achieve accurate segmentation by learning

feature representations from image to pixel level. The current mainstream segmentation framework primarily uses encoder-decoder structured networks, which predict pixel categories by abstracting and reconstructing features with the same resolution as the image. The fully convolutional network (FCN) performs convolution and deconvolution operations on the entire image, which recovers the abstract feature map generated after multiple convolution and pooling operations to the original size as the input image to generate segmentation results. This process achieves an end-to-end mapping from image to pixel prediction. The encoder-decoder structure captures contextual information along the encoder path while recovering high-resolution features in the decoder path. Both U-Net and SegNet adopt symmetric encoder-decoder structures. U-Net helps preserve spatial information and minimize information loss through skip connections linking the encoder and decoder paths. In contrast, SegNet uses max-pooling indices for upsampling, which helps recover segmentation details. Furthermore, the SPP network captures multiscale information by performing SPP at various grid scales. PSP-Net [138] and DeepLab [139], [140] exploit the multiscale information by using SPP [141], [142] at multiple grid scales or applying parallel atrous convolutions of different rates. DeepLab also involves depthwise separable convolution [143] and group convolution [144] to reduce computational cost and the number of parameters while maintaining comparable or slightly improved performance. For the OID tasks, researchers have optimized and improved each subfunction network within these segmentation frameworks. They have also introduced effective attention mechanism modules, loss functions, and training strategies to improve detection performance.

Regarding rail segmentation, most methods adopt an encoder-decoder structured segmentation framework, with U-Net [21] being predominant. Alexandrescu and Manole [145] used ResUNet++ [146] to introduce the ResNet structure into the U-Net framework for rail detection and adopted weighted binary cross-entropy loss as the loss function to improve rail detection accuracy. Sevi and Aydin [147] proposed an integrated segmentation network. The model uses U-Net as the base framework and integrates three networks, ResNet-34, MobileNetV2, and VGG16, into the backbone network to segment the rail through joint training. Gasparini et al. [148], [149] decomposed OID into three subtasks: anomaly detection, localization, and classification. They employed an AE network based on the U-Net framework for obstacle detection and localization, followed by obstacle classification based on the localization results. Jahan et al. [150] used a U-Net backbone to extract the rail from the background and then used an AE to divide the detected rail images into small patches for reconstruction. Obstacles are identified by comparing the differences between the original and reconstructed images. Based on SegNet, Wang et al. [151] optimized the network structure by adopting atrous convolution and cascaded multiscale features to improve the rail segmentation accuracy and implemented an improved polygon fitting strategy to achieve accurate detection of rail edges. Tao et al. [152] proposed a RailNet network that uses atrous convolution to reduce the

parameters in the U-Net structure and incorporates the channel attention module split soul (SS) to improve the accuracy of rail segmentation. The rail segmentation results were then combined with the inverse perspective transformation for rail polynomial fitting to accurately extract the rail. Belyaev et al. [153] applied SPP at the top feature layer based on the U-Net framework and introduced an attention refinement module (ARM) to perform feature fusion in the front and back channels, thereby improving the segmentation accuracy. Zhang et al. [154] proposed a DFA-UNet rail segmentation network, which integrates atrous convolution into the backbone network in the U-Net framework to improve feature extraction performance and constructs a global average maximum pooling (GAMP) channel attention module based on SE-Net to further improve the rail segmentation accuracy [155]. Inspired by SCNN [156], Li et al. [157] exploited the spatial geometric features of railway tracks to improve rail segmentation accuracy by incorporating horizontal and vertical attention modules into the output layer of the rail segmentation network. Lian et al. [158] introduced DMFNet, a lightweight and precise semantic segmentation framework for railway scenes, which addresses the challenge of semantic sample imbalance by using dense connections, multiscale fusion, and novel loss reallocation technique, achieving outstanding performance in real-time segmentation on railway datasets.

3) *AE- and GAN-Based Methods*: AE can automatically learn and extract compact representations of input unlabeled images through the encoder module and reconstruct the normal features of the image through the decoder, thereby facilitating unsupervised object recognition, while abnormal regions such as obstacles are difficult to recover. Gasparini et al. [148] proposed an AE that can reconstruct normal frames and classify whether there are anomalies in the frame by analyzing both absolute and gradient differences. Jahan et al. [150] proposed an AE to reconstruct nonanomalous rail scene images, which produces a significant reconstruction error for anomalous images compared to normal railway images. They determined the occurrence of obstacle intrusion by setting a threshold of reconstruction error. Boussik et al. [159] developed a rail OID algorithm based on AE convolutional network reconstruction. They introduced the concept of gap-score, which evaluated each model by calculating the average difference between the reconstruction scores of images with and without obstacles and identified the model that maximizes the average gap-score.

Different from CAE, the GAN incorporates a discriminator structure to evaluate the quality of image reconstruction, continuously optimizes parameter settings, and extracts more compact features to minimize the difference between the reconstructed image and the input image based on the results of the discriminator [160]. GAN trains a generator and a discriminator simultaneously. The generator aims to generate samples that resemble real data, while the discriminator attempts to distinguish between generated and real samples. This feature allows GANs to learn a mapping from random noise input to target output images, which is the key function in transforming an input image to a target output. GANs are particularly useful for addressing the challenge of obstacle data scarcity. Guo et al. [161] proposed a novel conditional deep

convolutional GAN (C-DCGAN) for OID. They used a public dataset to train a generative model to generate obstacle images, which were then scaled and fused with rail images based on visual perspective changes. Then, they detected obstacles using YOLOv3. Zheng and Liu [162] proposed a full-category obstacle detection model based on the fast diffusion model (DM). This model is trained exclusively on normal samples, and during detection, the DM partially reconstructs images with obstacles and reduces them to obstacle-free images. The reconstructed images are then compared with the original input images to identify intrusion objects, enabling simultaneous detection from multiple viewpoints. Similarly, Deng [163] proposed a GAN GANomaly to construct a generative model for obstacle detection by reconstructing image differences. Liu [164] improved the SRGAN GAN to enhance low-resolution images into high-resolution images, thereby improving image clarity and thus enhancing the obstacle detection performance of the SSD network.

4) *Few-Shot Learning Methods*: Typical deep neural networks in computer vision require large amounts of labeled training data to learn the latent data distribution and avoid overfitting. However, data collection and labeling can be time-consuming and labor-intensive, especially in railway applications. Effective training samples from an individual video camera can be limited since most video frames tend to be highly similar in the same railway scene. This leads to a computer vision scene characterized by small samples. Few-shot learning aims to achieve knowledge transfer by analyzing the similarities and differences between previous categories and new categories with very little data so that the trained model can accurately classify new categories. Gong et al. [165] proposed a meta-learning algorithm for detecting turnouts and crossings in disparate railway scenes and landforms, which requires less new data to accurately detect new environments. To enhance the meta-learner and alleviate the overfitting problem, various methods are employed, such as label shuffling, neuron dropout, and additional mask inputs extracted from the original video frames (of the training data) using FCN. Ye et al. [166] proposed an efficient few-shot object detection (FSRD) method for railway intrusion detection, which introduced a classification head initializer and comparative learning to improve detection accuracy and adaptability. The model was tested on a real railroad dataset and the results show that it outperforms the state-of-the-art methods in terms of redetection accuracy, adaptability, and generalization.

C. Track Trajectory Detection Methods

In addition to employing object detection and segmentation algorithms for rail detection, researchers have also explored ideas and methods from road lane detection to rail line detection. Generally, track trajectory detection methods can be divided into three categories: segmentation-based methods, parameter-based methods, and anchor-based methods. Among them, the anchor-based methods are further classified into row-anchor-based methods and line-anchor-based methods. The parameter-based methods mentioned in Section IV-A utilize predefined parameters or rules to model and regress for

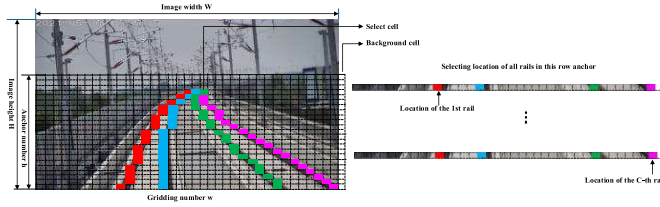


Fig. 7. Illustration of selected rail locations using the row-anchor-based method [90].

detecting rail lines [167]. However, they struggle with complex or occluded rail geometries and heavily rely on accurate initial point extraction. The pixel-level segmentation methods mentioned in Section IV-B2 utilize a pixel-level prediction method to convert rail lines into a multivalued mask; However, they are complex and computationally expensive, which may not be able to meet the requirements of real-time detection. Moreover, the accurate segmentation of the track depends on postprocessing, which may also lead to new deviations. This section focuses on row-anchor-based methods and line-anchor-based methods to accurately detect railway lines. Row-anchor-based methods adopt anchors and grindings to convert the mask into a series of horizontal locations. The primary strategy consists of predicting the horizontal coordinates of the railway lines, selecting the cell with the highest probability for each row, and repeating this process. Subsequently, postprocessing based on geometric algorithms is performed to reconstruct all discrete points into structured lines. In contrast, line-anchor-based methods employ predefined line anchors as references to accurately regress railway lines, thereby preserving the linear priori information of railways and integrating global information.

1) *Row-Anchor-Based Methods:* Row-anchor-based methods represent rail tracks as a series of horizontal positions corresponding to the intersections between predefined row anchors and the rail polylines. These positions are further discretized into class units through gridding, transforming the rail detection task into a row-wise selection problem, as illustrated in Fig. 7. In the final stage, all discrete points are reconstructed into structured rail lines via postprocessing. For example, Li and Peng [90] introduced an efficient row-anchor-based rail detection method, Rail-Net, which adopts a lightweight convolutional backbone, including ResNet with different layers, SqueezeNet, MobileNet, and Vision Transformer (ViT), alongside an anchor classifier. This method reduces computational cost compared to alternative segmentation methods. Ni et al. [168] proposed UFATD, a row-anchor-based track detection network that formulates railway track detection as a combination of perspective classification and row-wise selection. The network uses predefined anchor groups to grid the image, a perspective identifier to select the best group based on viewpoint, and a horizontal coordinate locator to predict track positions, enabling efficient and robust detection. Tong et al. [169] proposed TriRNet, a row-anchor-based real-time rail recognition network for unmanned aerial vehicle (UAV)-based railway inspection. The method introduces a general rail representation via anchor points on horizontal and vertical positioning lines, leveraging projection length

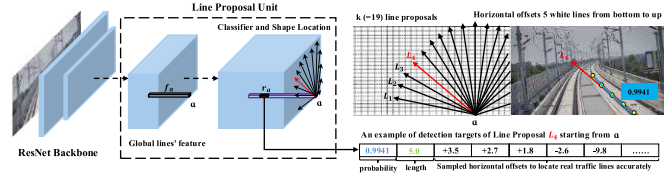


Fig. 8. Pipeline of the line-anchor-based rail track detection framework [170].

discrimination to adaptively choose the optimal representation direction. The proposed method achieved an F1@50 of 79.25% on a self-constructed railway dataset.

2) *Line-Anchor-Based Methods*: Line-anchor-based methods utilize predefined or dynamically generated line anchors to encode prior knowledge about the linear geometry of rails and to regress rail positions by leveraging these representations. For example, the Line-CNN [170] methods utilize predefined line anchors as references to accurately regress rail lines, aggregating global information while retaining the linear prior information of rails, as shown in Fig. 8. Zhou et al. [171] proposed an improved network, CLRNet-R, by introducing a lane detection method based on the similarity between rail and lane trajectories. This method incorporates an attention mechanism to enhance the global feature extraction capability and adds a semantic segmentation head to improve feature representation in the rail region. To address the problems of poor curve recognition and uneven output lines in the original CLRNet algorithm [167], the weight assignment for line IOU calculation within the framework is refined. Furthermore, two local slope-based loss functions are proposed to optimize the training constraints of the model’s local sampling points. This enhancement improves the model’s fitting performance for curved rails, leading to smoother and more stable rail line detection. Yu et al. [172] proposed DALNet, a rail detection method based on a dynamic anchor line (DAL) mechanism, addressing the limitations of predefined line-anchor-based approaches. DALNet generates anchor lines dynamically for each rail instance using predicted starting points and slopes. This improves alignment with actual rails and reduces redundant predictions without requiring NMS.

D. Camera-Based Distance Estimation Methods

In addition to rail and obstacle detection, some researchers have explored methods to estimate obstacle distances. Monocular vision methods rely on a single camera and thus cannot directly obtain depth; they typically use geometric assumptions or prior knowledge, such as vanishing points, known object sizes, or image-plane homography, to infer distance. For example, Huang et al. [173] proposed FRVO-Mono, a monocular visual odometry method for railway localization, that uses vanishing points, track lines, and pole objects to provide geometric constraints. The method addresses scale drift and improves object matching accuracy, achieving meter-level accuracy by restoring absolute scale using camera height, with potential for submeter-level positioning when combined with additional semantic information. Haseeb et al. [78] and Ristić-Durrant et al. [79] proposed a technique for estimating the distance from a monocular camera to an observed object. They first employed the YOLO network to detect obstacles within the rail region

of the image. They then built a multilayer neural network that leverages object features along with the physical distance between the object bounding boxes and actual measurements of the distance to the obstacle. This approach was implemented in the SMART project [10]. Stereo vision and multicamera systems alleviate monocular limitations by capturing spatial information from multiple viewpoints, enabling more accurate depth estimation. Building on Haseeb's work, Abdul et al. [81] fused the detection results of two cameras (color and IR) to develop a multilayer neural network for predicting object features and distances (0–1000 m), addressing limitations in harsh weather. Wang and Wang [174] used the self-attention distillation (SAD) algorithm to detect rails and the YOLOv4 network to detect obstacles in the image. They then mapped the detection results to millimeter-wave radar data to obtain the real-world distance of the obstacle. Ćirić et al. [175] developed the FuzzyH method, integrating thermal camera imaging with image-plane homography and fuzzy logic to handle uncertainty and nonlinearity in distance estimation under low-light conditions. Tested on Serbian railways, FuzzyH reduced maximum error from 90 (13.5%) to 32 m (7.1%) and maintained <5% average error over 400–950 m, demonstrating robustness against poor lighting and off-center camera angles.

E. Sensor Fusion and Event-Based Vision Methods

Although most existing approaches rely on conventional cameras equipped with CCD or CMOS sensors, some studies have explored sensor fusion methods that combine LiDAR point clouds or other sensor data with image data, as well as the use of event-based cameras to address the challenges of perception in low-light and high-speed dynamic environments.

Considering the object recognition capability, current OIDS primarily rely on data captured from cameras and LiDAR. However, integrating these heterogeneous modalities presents significant challenges, such as spatial misalignment, temporal asynchrony, and feature incompatibility. To address these issues, three primary strategies for multimodal fusion have been widely adopted: data-level fusion, feature-level fusion, and decision-level fusion. Data-level fusion directly combines raw inputs based on geometric calibration or semantic projection. A typical example is PointPainting, which overlays image-based semantic segmentation masks onto LiDAR point clouds. Although conceptually straightforward, this strategy requires precise calibration and incurs high computational overhead. In railway-specific applications, Karaduman [176] proposed a multisensor obstacle detection method that combines image template matching with laser measurements at the data or early fusion level to reliably detect obstacles around turnouts. Feature-level fusion operates on modality-specific features extracted from cameras and LiDAR, which are then integrated at an intermediate stage to exploit cross-modal complementarities. Wang et al. [177] introduced FarNet, a point cloud segmentation network leveraging spherical projection to convert the LiDAR point cloud into a pseudo-distance image and attention aggregation for rail segmentation, enhancing detection precision for obstacle identification, as shown in

Fig. 9. Yang et al. [178] proposed RailDepth, a self-supervised depth completion framework fusing sparse LiDAR and RGB images using a novel fusion module with pooling-guided attention to improve depth estimation in challenging rail transit environments. Cao et al. [179] developed a robust multimodal obstacle detection method that reconstructs rail topology by fusing RGB and LiDAR features, which not only enables accurate detection of obstacles located on or near the rail tracks but also improves the spatial consistency and context awareness of the detection process. In addition, Bai et al. [180] proposed a lightweight camera–LiDAR fusion approach for rail transit, which performs feature-level fusion by first extracting modality-specific features using lightweight CNN backbones, followed by projection into a shared BEV space to ensure spatial alignment across sensors. These examples reflect the advantages of feature-level fusion in producing expressive, complementary representations. Decision-level fusion, in contrast, combines the final outputs from independent 2-D and 3-D detectors to improve robustness and efficiency. Assaf et al. [5] developed an auto-adjustable camera–LiDAR system that uses camera data to locate ROI, enabling LiDAR to focus obstacle detection efforts and fuse the resulting decisions for reliable intrusion identification. Similarly, Kageyama et al. [181] demonstrated a train frontal obstacle detection approach employing decision-level fusion to combine camera and LiDAR detections for improved accuracy at long range. These fusion paradigms have been effectively applied to enhance robustness, precision, and adaptability of OIDS in diverse operational scenarios.

Event cameras differ from traditional frame-based cameras by asynchronously outputting pixel-level events only when brightness changes occur, enabling microsecond-level temporal resolution and reduced data redundancy. Due to their high sensitivity to light changes, event cameras are effective in low-light environments and have been applied to intrusion detection by encoding event streams for neural network processing. Gebler and Pohle-Fröhlich [94] proposed a method for intrusion detection at railway tunnel entrances using a dynamic vision sensor (DVS). The event stream generated by the DVS is encoded into either a time surface or an event volume, which is then fed into several neural networks for classification. The method was evaluated on a custom dataset, achieving an accuracy of 99.55% with MobileNetV2 on Event Volumes, addressing challenges such as extreme lighting and high-speed train turbulence. Walters and Hadfield [182] integrated event data with LiDAR measurements to estimate metric time-to-impact, eliminating the need for prior information about scene geometry or obstacles. Tschopp et al. [183] proposed Hough2Map, an iterative event-based Hough transform framework for high-speed railway mapping. Leveraging a DVS to address motion blur and illumination challenges, the method detects vertical landmarks (e.g., power line poles) by encoding event streams into a Hough space with constrained angles. Experiments on real-world tram datasets demonstrate 92% detection reliability and a mapping root mean square error of 1.1518 m, validating its efficiency for railway localization without external infrastructure.

V. KEY CHALLENGES AND FUTURE BREAKTHROUGHS FOR RAILWAY OIDS

In recent years, with the development of advanced sensors and the integration of cutting-edge AI technology, research on rail OID has made significant progress. OIDS has been used in rail sections, greatly promoting the rapid automation and intelligence of railway inspection. However, in complex operational environments such as extreme weather, mountainous areas, and tunnels, the technique still faces some challenges and urgently needs to meet requirements such as high real-time response, low false alarm rates, and limited hardware computing resources. While researchers have explored solutions using DL methods, these approaches still have limitations in terms of visual features and model performance. Challenges include misdetection or omission of small-scale obstacles due to low pixel occupancy of distant or small objects in the image. In addition, the effectiveness, generalization, and robustness of algorithms are hindered by the difficulty in obtaining diverse images of railway scenes and potential obstacles under various conditions. There is also a conflict between the large-scale parameters generated by model training and the need for efficient model deployment. This section summarizes three potential research directions for further development of obstacle detection: small-scale object detection, generating high-quality samples for specific scenes, and achieving a balance between a large number of parameters and efficient deployment.

A. Small-Scale Object Detection

Different obstacles vary in size, and the perspective projection from real-world scenes onto the image plane alters the scales of objects. The size of the same object can be different in different frames. Small-scale objects, such as rocks, sheep, and children, are common in rail scenes. Compared with objects of traditional sizes, small-scale objects occupy fewer pixels in the image and offer less information, resulting in degraded detection performance. In recent years, improvements in DL methods have significantly ameliorated the performance of small object detection. Techniques have been devised to integrate features across layers, enhance the available information at each layer, balance positive and negative samples of small objects, and increase the number of small object instances [184]. Three effective approaches to address the challenges of small-scale detection include multiscale feature fusion and context-based methods to enhance feature relationships, GAN- and DM-based methods to enrich the information of small objects, and data augmentation techniques to balance the distribution of objects of different scales.

1) *Multiscale Feature Fusion and Context-Based Methods:* During the feature extraction process in DL models, the spatial information of small objects may be lost due to the convolution calculation. One of the solutions is to integrate feature maps from both shallow and deep layers to combine spatial and semantic information. To achieve better feature fusion, various feature fusion schemes have been devised and developed, such as FPN [26], PANet [27], NASFPN [33], and bidirectional concatenation (BiC) module [185]. Building on

these approaches, Bi-FPN [28] and the adaptive spatial feature fusion (ASFF) module [186] can automatically learn fusion weights according to the importance of each feature, so that the network can maintain consistency across different feature scales. Attention models based on the GCA mechanism, such as the squeeze-excitation network (SENet) [155], selective kernel network (SKNet) [187], and ResNeSt [188], have shown great potential in feature fusion by emphasizing valuable regions and ignoring irrelevant parts. Based on this concept, the multiscale channel attention module (MS-CAM) [189] aggregates multiscale contextual information along the channel dimension to achieve cross-layer feature fusion. By utilizing the spatial attention mechanism, 1-D global pooling in the horizontal dimension can capture global information in the vertical direction, while 1-D global pooling in the vertical dimension can capture global information in the horizontal direction, which is conducive to the recognition of the linear structure of the rail.

Compared with large- or medium-scale objects, small-scale objects usually have lower resolution, resulting in limited visual information and challenging to extract discriminative features after multiple layers of convolution. Contextual information representation is crucial for small object detection. Local context is typically exploited by widening the detection window [190], [191]. Inspired by vanishing point-guided lane detection [192], Chen et al. [193] proposed a vanishing point-guided context-aware network (VCANet) to address the challenge of detecting small dangerous obstacles on the road from a distance. Inspired by the human ability to use contextual information in the vertical direction of the view to identify traffic signs, Yuan et al. [194] developed a vertical spatial sequence attention network (VSSA-Net) to explicitly encode vertical spatial context for more accurate traffic sign classification. Yuan et al. [195] proposed an adaptive spatial-temporal context-aware (ASTCA) model that incorporates spatial context information into the DCF-based tracker for UAV-tracking tasks, which alleviates background interference and adapts to variable viewpoints. Ni et al. [196] proposed a two-stage Transformer-CNN-based framework for multitype rail segmentation, in which the attention-based superpixel segmentation subnetwork is used to fit rail boundaries, and the superpixel-guided multitype rail segmentation subnetwork utilizes global context and structured priors to aggregate rail features to make segmentation decisions at both superpixel and pixel scales.

2) *GAN-Based and DM-Based Methods:* Super-resolution representation is crucial for small object detection. The GAN-based methods [197] enrich the information of small objects by increasing the resolution of the input image, effectively mapping low-resolution small object features to high-resolution features. This minimizes the feature gap between small and large objects, making the detection performance comparable to medium and large objects. Li et al. [198] proposed a perceptual GAN method specifically for small object detection within the GAN framework. This method aims to improve the feature resolution of small objects through mutual adversarial learning between the generator and discriminator. Noh et al. [199] proposed a feature-level super-resolution technique that

employs an SR target extractor with inception convolution [200] to effectively improve resolution while preserving the relative perceptual field. Yuan et al. [201] applied a CNN model inspired by SRCNN to improve the spatial resolution of benchmark hyperspectral images. Courtrai et al. [202] refined the super-resolution framework based on CycleGAN [203] by replacing the generator of Wasserstein GAN [204] with EDSR [205] residual blocks for small object detection in aerial remote sensing images. Based on the above, super-resolution GANs EESRGAN [206], EEGAN [207], and ESRGAN [208] have been proposed for small object detection based on edge enhancement. While GANs represent the state-of-the-art in many image generation tasks, they often struggle to capture less diversity and are difficult to train, limiting their scalability and applicability to new domains. In contrast, DMs are a class of likelihood-based models that have recently demonstrated the ability to produce high-quality images. Dhariwal and Nichol [209] found that DMs can achieve image sample quality superior to current state-of-the-art generative models, making the development of small object detection algorithms based on DMs a promising area for further research. Zhang et al. [210] proposed the VerseDiff-UNet network, which integrates the denoising diffusion probabilistic model (DDPM) into the standard U-shaped architecture. This method combines noise-added images with labeled masks to guide accurate diffusion direction, addressing the challenges associated with high noise, ambiguity, and uncertainty. The DM can fit the distribution of background images and thus isolate small targets from the distribution [211].

3) *Data Augmentation Methods*: Some datasets contain a limited number of images with small objects, or each image may contain too many instances, leading to a suboptimal distribution of small objects. Several studies have addressed this issue from the dataset perspective. Kisantal et al. [212] used an oversampling strategy to duplicate images containing small instances, addressing the problem of insufficient small object images. They also copied and pasted small targets into multiple locations in the image to enhance the diversity of small object positions and alleviate the problem of insufficient anchor boxes for matching small objects. However, replication augmentation methods that increase the number of small objects by randomly changing the brightness and size of objects and blending them into the background may result in background and scale mismatches [213]. To address these issues, Chen et al. [214] adopted a pretrained semantic segmentation network in their proposed RRNet. This approach places the replicated instances in a position that matches their backgrounds and uses a linear function to calculate the proper scale, leading to better results.

B. High-Quality Sample Generation

Although OIDS have been deployed on actual rail sections and the corresponding rail scene datasets have been established, there are still some challenges affecting the efficiency, robustness, and accuracy of the system. A major issue is the limited availability of diverse samples, leading to under-representative training datasets. It is difficult to capture rail scenes under all operating conditions, especially those involving potential obstacles. Another challenge is model

specificity. The generalization and robustness of recognition under different environments are usually low because the rail scenes captured by cameras in varied terrains have significant dissimilarity. Conversely, video frames captured by a single camera in a consistent background tend to have high similarity. To address these challenges, three effective approaches can be considered: few-shot learning using small sample data, fine-tuning large (foundation) models using small sample data, and building virtual physical models to generate the sample data required for training.

1) *Few-Shot Learning*: For object intrusion detection in railway environments, where abnormal classes like pedestrians or track obstructions are rare, few-shot learning plays a critical role in generating diverse and discriminative samples for model training under data-scarce conditions. Few-shot learning facilitates information transfer by focusing on the similarities between previously learned classes and the current class and reusing model parameters. It can generally be classified into three categories: data augmentation, metric learning, and meta-learning.

Data augmentation techniques aim to increase training data and generate diverse data under limited training through techniques such as image transformation, synthetic sample generation, data mixing, feature space augmentation, and pseudo-sample generation. Yu [215] proposed a two-stage few-shot detection framework for railway overhead line intrusions and constructed an enhanced railway dataset through image augmentation and scene transfer. It detected rare railway-specific objects by combining pretraining on common classes with fine-tuning on a balanced dataset. While classical methods such as geometric transformations and pixel-level noise addition may improve low-level diversity, they often fail to generate semantically meaningful variation. In contrast, recent generative approaches [216], [217], such as GAN- or diffusion-based augmentation, have demonstrated superior performance by producing context-consistent and class-specific synthetic samples. However, these models often require extensive pre-training and suffer from domain gap issues, particularly when applied to safety-critical domains like railway intrusion detection.

Metric learning focuses on learning a distance metric function that minimizes the positive sample distance (intra-class) and maximizes the negative sample distance (inter-class). Wang et al. [218] showed that connecting a deep model with well-designed metric objectives can effectively capture sample similarity. Duan et al. [219] introduced deep adversarial metric learning (DAML), which creates difficult-to-score negative samples from identified negative samples, where the potentially hard negative samples serve as a supplement for further discrimination. Hard negative mining is particularly useful in detecting visually similar intrusions. Ye et al. [166] proposed an efficient FSRD method for railway intrusion, combining fine-tuning and contrastive learning. It applied a contrastive loss to reduce intra-class variance and strengthen class separation. The method achieves 80.51% mAP50 under a 30-shot setting on the RT2022 dataset. Some studies [220], [221], [222] also proposed customized loss functions to make similar samples close together and dissimilar samples far apart or to

minimize the distance between the anchor and positive samples and maximize the distance between the anchor and negative samples.

Meta-learning aims to generalize across tasks by learning from multiple related tasks. Some studies [223], [224] utilized the attribute knowledge between different categories to apply small sample meta-learning methods to structural damage identification or utilized the uncertainty integrated within the meta-learning framework to improve the robustness of road scene anomaly detection in autonomous driving. Kim et al. [225] explored hybrid frameworks that combine few-shot learning with pseudo-labeling to enhance sample quality and class discrimination. In addition, integrating with uncertainty estimation has been shown to be effective in improving robustness against occlusion and low-resolution input.

2) *Fine-Tuning of Large (Foundation) Models*: Foundation models refer to large-scale AI models pretrained on diverse and massive datasets and then fine-tuned for a specific task. The introduction of the Transformer architecture facilitates the training of models with billions or even trillions of learnable parameters across a wide variety of data types and enables transfer to a wide range of downstream tasks. Large training datasets, including those containing hundreds of millions of images for computer vision tasks, are also available [226]. Foundation models are typically task-agnostic, trained on annotated datasets, and serve as a foundational component for generative AI [227]. Once pretrained, these models can be fine-tuned to tackle a wide range of specialized tasks using relatively small amounts of task-specific annotated data [228], [229], significantly reducing the need for extensive task-specific datasets.

Foundation models are often trained using self-supervised learning and generative methods, where the supervisory signals come from the context of the input data, thereby eliminating the need for manual annotation. These training paradigms allow foundation models to generalize well across tasks and domains, supporting them to perform zero-shot and few-shot inference. Once pretrained, foundation models can be fine-tuned with relatively small amounts of task-specific labeled data. This few-shot fine-tuning capability is particularly relevant to OIDS applications, where rare obstacle categories (e.g., fallen trees and foreign objects) are inherently difficult to capture in large numbers. For example, models such as CLIP [230] and ALIGN [231] have demonstrated that language supervision can play an important role in pre-training vision–language models, enabling them to perform various visual recognition tasks through zero-shot learning. For instance, CLIP can be prompted with “detect a tree fallen across the railway track,” allowing for zero-shot detection of previously unseen intrusions. Similarly, multimodal generative models such as Stable Diffusion [232] or DALL-E demonstrate the capability to synthesize realistic, domain-specific visual scenarios from textual descriptions, including unusual weather conditions and complex railway environments. These generated images can augment limited training datasets in OIDS applications and enhance model robustness to diverse conditions. Generative foundation models learn the joint or conditional probability distribution over the training input data,

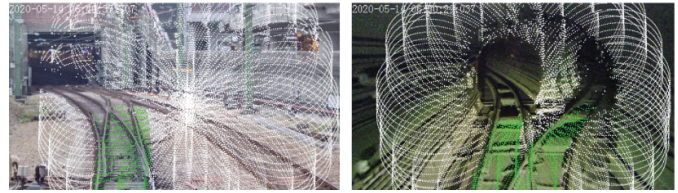


Fig. 9. Rail segmentation results in turnout scenarios based on LiDAR [177].



Fig. 10. Train simulator editor [240].

allowing them to accurately generate patterns resembling the input data used during training. Wang et al. [233] showed that advanced image synthesis models like DALL-E can create designs that are indistinguishable from those made by human designers. These text-to-image foundation models not only support zero-shot creativity but also serve as effective tools for augmenting training data in OIDS. Recent trends in text-to-video (T2V) synthesis further extend the range of multimodal generation for virtual simulation and intelligent monitoring systems. Cong et al. [234] proposed FLATTEN, a novel training-free framework for T2V editing that introduces optical flow-guided attention into the diffusion process. By estimating patch trajectories from the source video via optical flow and enforcing attention only among patches on the same trajectory, the method solved the problem of poor temporal consistency in video editing.

3) *Virtual Reality Technology and Generative Models*: To overcome the scarcity and limited diversity of real-world railway scene data, especially under rare or dangerous conditions, high-fidelity simulation engines combined with generative modeling have emerged as a powerful solution. A simulation platform can build a realistic railway environment and allow the insertion of various obstacles, for instance, pedestrians, animals, or falling rocks. By leveraging physics-based rendering and procedural generation, virtual environments can simulate intrusion events with high photorealism and physical plausibility, which are difficult to replicate in the real world. This enables not only high diversity and morphological randomness of obstacles but also the simulation of various environmental conditions (e.g., fog, rain, and low lighting). Virtual cameras embedded in these environments capture images that are automatically annotated with spatial and semantic labels, thereby reducing reliance on manual labeling and significantly enriching the training datasets used

for OIDS. For example, some studies [235], [236], [237], [238] have considered extracting pixel-level semantic label maps from computer games to supplement real-world images or superimposing 3-D physical models onto real images in virtual environments to customize target objects. Some railway technology suppliers, like CAF signaling, have started exploring computer vision and AI (CV&AI)-enhanced technologies for artificial sensing of the environment. Pan et al. [239] proposed a virtual railway scene data generation framework based on Unreal Engine. The system leverages real-time rendering and spline mesh components to build tracks and control train trajectories, with a camera mounted on the locomotive to capture dynamic scenes. By incorporating terrain, vegetation, and human/animal models, the system creates diverse obstacle scenarios. This method enhances data diversity and model generalization under extreme conditions while avoiding the safety risks of real-world data collection. Labayen et al. [240] proposed a semi-automatic framework for validating CV&AI-based railway systems using virtual scenarios, as shown in Fig. 10. Their approach uses virtual scenarios to simulate various visibility conditions and diverse scenarios that would be difficult and expensive to capture in real environments, improving test quantity, quality, and cost efficiency. Their work lays the foundation for fully automated validation and verification systems, potentially integrating open-source tools like CARLA. D'Amico et al. [100] proposed RailSim, a simulation framework proposed RailSim, a simulation framework built on Unreal Engine 5 to generate photorealistic virtual railway environments for testing visual odometry algorithms. The approach integrates railway from OpenStreetMap to construct diverse scenarios (urban, farmland, and industrial areas) and simulates sensors (cameras and LiDAR) to capture synthetic data. By applying the ORB-SLAM2 algorithm to both RailSim-generated synthetic images and real-world data from the OSDaR23 dataset, the study validates the feasibility of virtual simulation in the early testing of railway perception tasks. Han et al. [241] proposed an artificial neural network (ANN)-based high-speed railway virtual reality 3-D simulation scene optimization model to improve rendering efficiency.

Simulation engines are often constrained by the fixed asset libraries and may lack full control over physical parameters. In contrast, a physics engine provides greater flexibility to simulate precise structural deformations or weather dynamics. For example, physics-based graphics modeling (PBGm) [242] was built to generate 3-D object graphical representations of target structures and the associated environment and to simulate the entire data collection on a computer. The same method has also been used to synthesize ideal environments, such as the synthetic environment for post-earthquake inspection of high-speed railway viaducts [243], the synthetic environment for reinforced concrete railway viaducts [244], and the synthetic environment for track obstacle intrusion [101]. For example, Tulembayeva et al. [245] proposed an automated flight algorithm for UAVs. The approach employs Webots R2023b simulator (based on open dynamics engine (ODE)) to build a 3-D virtual environment to develop the UAV's autonomous flight control, initializing its GPS, camera, and modules to model and test the automatic flight algorithm of the

drone, including simulation of railway track recognition and train detection. Moreover, digital twin models, which integrate physical entities, virtual entities, and their interactions, enable 3-D dynamic perception of infrastructure damage states with a higher degree of simulation [246].

However, the realism of synthetic data often hinges on the fidelity of the graphics engine and physics simulation, which may still leave a domain gap between virtual and real data. To mitigate this issue, domain adaptation techniques such as style transfer (e.g., CycleGAN), domain-adversarial neural networks (DANNs), and domain-invariant feature learning are applied to bridge the visual and statistical differences, like in [95] and [202]. Moreover, advanced generative models, like Stable Diffusion, could provide controllable, high-fidelity generation that allows fine-grained simulation of scene elements, further enhancing realism and domain transferability, like in [209], [210], and [234]. Furthermore, Guo et al. [247] proposed a method leveraging DMs (Stable Diffusion) to generate rare and hard-to-capture obstacles. The approach designs prompt templates with variables like weather and motion to drive Stable Diffusion, extracts obstacle masks using SAM, and pastes them onto real railway backgrounds with image harmonization. This DM-assisted pipeline enables efficient generation of diverse obstacles without manual annotations, filling the gap of existing railway datasets and improving zero-shot detection performance.

C. Conflict Between a Large Number of Parameters and Efficient Deployment

Deploying DL models in railway OIDS requires a careful balance between accuracy and efficiency. Comprehensive models trained on diverse and complex railway scenes tend to be large with millions or even billions of parameters, posing significant deployment challenges. Shallow neural networks designed for railway scenarios often exhibit high specificity but low generalizability due to the limited number of parameters and insufficient computing power, which hinders the improvement of detection accuracy. In contrast, trained on large-scale, high-quality rail and obstacle datasets, DL frameworks such as CNN, recurrent neural network (RNN), and Transformer have demonstrated superior performance on obstacle detection tasks. However, their massive parameter counts pose challenges in memory, latency, and real-time deployment. These limitations are particularly critical in edge computing environments such as onboard systems or track-side inspection units, where power, latency, and storage are constrained.

To facilitate the rapid deployment of DL models in real-world railway scenarios, three strategies have been widely adopted: 1) designing lightweight model architectures, such as MobileNet, YOLOv5-Nano, and EfficientNet-Lite, which are inherently efficient; 2) applying model compression techniques, such as pruning, quantization, knowledge distillation (KD), and low-rank decomposition, to reduce the computational demands of pretrained large models; and 3) deploying models directly on edge devices to support real-time inference in field applications. These approaches aim to compress or adapt complex models to compact ones and

optimize architectural design, enabling deployment in real-world, latency-sensitive railway systems.

1) *Lightweight Model Design*: Lightweight model architectures are designed to achieve high efficiency with fewer parameters and lower computational demands. As discussed in Section IV-B [132], [133], [134], [135], [136], [137], [158], various studies have adopted lightweight network designs for railway OID tasks. These methods involve replacing the original YOLO or Transformer backbone with more efficient lightweight networks, employing lightweight convolutional modules to reduce parameters and computational cost, or integrating lightweight components to compress the original model. Such designs enable real-time inference on mobile or embedded OIDS platforms while maintaining satisfactory detection performance.

2) *Model Compression Techniques*: While lightweight model architectures are efficient and suitable for deployment on resource-constrained platforms, they may suffer from limited representational capacity and reduced generalization performance in complex environments. In contrast, model compression techniques, such as pruning, quantization, and KD, can leverage the expressive power of large models and transfer knowledge to smaller ones, offering better performance tradeoffs without redesigning the model architecture from scratch.

Large-scale parameters limit the deployment of DL models in OIDS. KD is a widely adopted compression technique for enabling lightweight and real-time object detection in OIDS. It transfers the knowledge from a large, high-performing “teacher” model to a smaller “student” model, allowing the student to approximate the teacher’s accuracy while significantly reducing computational costs. This improves inference speed and facilitates efficient deployment on resource-constrained platforms such as onboard systems. Meng et al. [248] proposed a fast railway foreign object intrusion detection method based on cascaded convolution neural network and overhaul KD. A lightweight network classifies images in the first stage, followed by YOLOv3 for object detection in intruded images. Compared with other approaches, this method improves efficiency and achieves competitive accuracy with a significant increase in FPS. Evain et al. [249] used dataset combination and KD to create lightweight models to improve the efficiency of 3-D monocular object detection for autonomous vehicles and railway systems. Shen et al. [250] proposed an unsupervised railway intrusion detection method called heterogeneous uninformed students network (HUS-Net), which uses a pretrained description model as the teacher network and distills its knowledge into two heterogeneous students. The representation difference between the students and the teacher was exploited to identify railway intrusions and locate abnormal objects.

Model pruning, NAS, and quantization are also widely adopted model compression techniques that significantly reduce the computational and memory footprint of deep neural networks. Pruning eliminates redundant weights or channels, either in an unstructured or structured manner, enabling efficient inference while preserving model accuracy. Guo et al. [251] proposed an improved SSD for high-speed railway

boundary intrusion detection. It utilized a recursive $L1$ -norm pruning strategy with arithmetic and geometric operations to compress convolution kernels layer-wise, reducing processing time to 31 ms/frame on GTX1060 with <1% accuracy loss while improving mAP by 2.52% and 4.74% on PASCAL VOC and railway datasets, respectively. Jiang et al. [252] introduced a lightweight traffic sign detection algorithm based on a hybrid pruning strategy combining channel pruning and weight pruning, that compresses the model to 12.4 MB (86% pruning rate) with 90.58% mAP and 142.2 FPS on the TT100K dataset, making it suitable for edge deployment. Quantization reduces the numerical precision of weights and activations (e.g., from 32-bit float to 8-bit integer), often with minimal impact on detection performance. Chenglin et al. [253] proposed an edge-enabled real-time railway track segmentation method based on YOLOv8-seg, which employs post-training quantization to convert FP32 model weights to FP16, reducing computational load and memory footprint. NAS aims to search the reasonable network in limited resources. Wang et al. [254] developed EANet, an NAS method based on evolutionary algorithms. It designs a search space with shared dilated convolutions to integrate traditional and depth-wise separable convolutions, and then, it encodes convolutional channel counts as genes and uses SENet-based variance analysis to optimize architectures. The resulting lightweight SSD backbone achieves both high detection accuracy and model compression for railway intrusion detection. Wang and Yu [255] proposed DRENet, an NAS method integrating dynamic receptive field, measurable dense residual connections, and SENet into the MobileNetV2-based search space. It adaptively optimizes network architecture via learnable parameters for kernel scaling and cross-layer connections, modeling costs as regularization loss to balance accuracy and efficiency. The DRENet enhances small-object detection in railway surveillance. Besides their application in OID, these techniques have been extensively explored in related fields such as autonomous driving [256], drone surveillance [257], and traffic monitoring [258], offering insights for railway OIDS deployment.

3) *Edge Device Deployment*: In practice, lightweight network designs and model compression techniques are often used in combination to meet the stringent computational and latency constraints of real-world deployment. Although deploying large-scale DL models on devices remains challenging, edge computing deployment provides a practical means to alleviate the tradeoff between model complexity and deployment efficiency. Combined with model compression techniques such as pruning, quantization, and KD, it supports the deployment of compact yet accurate models for real-time railway object detection tasks.

Deploying computing resources at the network edge helps to address key limitations of traditional cloud-based architectures, such as variable latency, overuse of the communication network [259]. As a result, edge computing deployment has emerged as a preferred paradigm for latency-sensitive services that demand high data processing and real-time contextual adaptation. This is especially crucial in railway systems, especially beneficial in remote or bandwidth-constrained environments, where timely response and local decision-making

are critical. Zhang et al. [260] implemented an edge computing architecture using Jetson Xavier for railway obstacle detection. By running YOLOv5 with ASFF/GAM and processing multi-channel 720-pixel video via DeepStream, the system improved mAP@0.5 by 2.5%, maintained real-time performance, and efficiently offloaded tasks from Jetson Nano to reduce latency. Tulembayeva et al. [245] proposed an edge-integrated UAV system for railway reconnaissance. A Raspberry Pi 4 running YOLOv4-tiny processes real-time video from a DJI MAVIC 2 PRO, extracting key data such as GPS and carriage count locally. Results are transmitted to ThingSpeak at only 13 KB/s, minimizing bandwidth and server load. Amin et al. [261] introduced an edge-cloud framework for grade crossing safety, combining weighted box fusion ensembles of YOLOv8M/L/X (97% precision) with a UNet-EfficientNetV2B3 segmentation model (98% IoU). Real-time video is processed on edge servers at 10 FPS and 1-ms latency, with only critical outputs sent to the cloud for retraining and storage. The edge-cloud architecture optimizes real-time responsiveness and leverages cloud resources for model retraining, ensuring both low-latency safety. Humes et al. [262] proposed Squeezed Edge YOLO, a hardware-efficient detection model tailored for edge deployment. By shrinking input size to 128×128 , replacing 3×3 with 1×1 convolutions, and removing residual blocks and a detection head, the model reduces weights by 90%. It is quantized to 8-bit for GAP8 deployment, achieving 130-ms latency, $3.3\times$ higher throughput, and 76% energy savings on Jetson Nano. Chen et al. [253] introduced a YOLOv8-seg-based real-time rail segmentation model optimized for edge devices. It integrates Ghost convolution, a lightweight head, and FP32-to-FP16 post-training quantization, achieving 83.3% accuracy at 25 FPS on Jetson Nano, with 3.0-MB model size and 37.5% reduction in GFLOPs. Overall, the deployment of OIDS models on edge devices provides a practical foundation for achieving reliable, low-latency, and scalable solutions in diverse and challenging railway environments.

VI. CONCLUSION

The sensing devices of the current OIDS are mainly based on cameras, supplemented by LiDAR and other sensors, which collect high-quality rail scene data. The OID scheme is primarily based on computer vision algorithms, which fully extract and learn image features to automatically detect rail and obstacles instead of manual inspection. Moreover, OID methods based on computer vision have evolved from traditional image processing methods that manually extract low-level features to DL methods that automatically extract high-level features and use different network architectures according to the image features of tracks and obstacles. Furthermore, improved algorithms are designed to address scene-specific challenges in railway environments, ultimately enabling efficient real-world deployment.

From a methodological perspective, this article systematically describes the structure, characteristics, advantages, and limitations of typical algorithms based on computer vision technology, as well as network structure optimization and improvement methods based on OID tasks, and an introduction to methods such as modules, loss functions, and training

strategies. Aiming at the three major challenges faced by obstacle detection systems when deployed in actual operation sections: small-scale object detection, lack of high-quality data for specific scenarios, and the contradiction between a large number of parameters and efficient deployment, several potential strategies are proposed: 1) using feature fusion methods and generative models for small object detection; 2) using few-shot learning and foundation model fine-tuning methods based on limited data to expand the dataset; 3) establishing physical virtual models to generate sample data and customized scenarios required for training; and 4) utilizing lightweight network design and model compression methods to achieve rapid deployment of large-parameter models. Finally, OIDSs achieve real-time detection on the vehicle and track side. In summary, the research on rail OIDSs has significantly promoted the rapid development of rail transit automation and intelligent detection technology by equipping and deploying advanced sensors and combining them with innovative computer vision methods and improvement strategies.

REFERENCES

- [1] Y. Qu, B. Wang, R. Qiu, and C. Zheng, "The research and realization of contactless collision early warning assistant system," in *Proc. Int. Conf. Electr. Inf. Technol. Rail Transp.*, 2016, pp. 147–154.
- [2] L. Guan, X. Li, H. Yang, and L. Jia, "A visual saliency based railway intrusion detection method by UAV remote sensing image," in *Proc. Int. Conf. Sens., Diag., Prognostics, Control (SDPC)*, Beijing, China, Aug. 2020, pp. 291–295.
- [3] A. Mammeri, A. J. Siddiqui, and Y. Zhao, "UAV-assisted railway track segmentation based on convolutional neural networks," in *Proc. IEEE 93rd Veh. Technol. Conf. (VTC-Spring)*, Apr. 2021, pp. 1–7.
- [4] D. Ristić-Durrant, M. Franke, and K. Michels, "A review of vision-based on-board obstacle detection and distance estimation in railways," *Sensors*, vol. 21, no. 10, pp. 3452–3483, May 2021.
- [5] E. H. Assaf, C. von Einem, C. Cadena, R. Siegwart, and F. Tschopp, "High-precision low-cost gimballing platform for long-range railway obstacle detection," *Sensors*, vol. 22, no. 2, p. 474, Jan. 2022.
- [6] D. He, Z. Zou, Y. Chen, B. Liu, and J. Miao, "Rail transit obstacle detection based on improved CNN," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–14, 2021.
- [7] M. Adham, M. G. Mohamed, and A. El Shazly, "Railway tracks detection of railways based on computer vision technique and GNSS data," in *Proc. ICCSTE*, 2020, pp. 1–8.
- [8] M. Ghorbanalivakili, J. Kang, G. Sohn, D. Beach, and V. Marin, "TPE-Net: Track point extraction and association network for rail path proposal generation," in *Proc. IEEE 19th Int. Conf. Autom. Sci. Eng. (CASE)*, Aug. 2023, pp. 1–7.
- [9] H. Salmene, L. Khoudour, and Y. Ruichek, "A video-analysis-based railway-road safety system for detecting hazard situations at level crossings," *IEEE Trans. Intell. Transp. Syst.*, vol. 16, no. 2, pp. 596–609, Apr. 2015.
- [10] D. Ristić-Durrant, M. A. Haseeb, M. Banić, D. Stamenković, M. Simonović, and D. Nikolić, "SMART on-board multi-sensor obstacle detection system for improvement of rail transport safety," *Proc. Inst. Mech. Eng., F, J. Rail Rapid Transit*, vol. 236, no. 6, pp. 623–636, Jul. 2022.
- [11] D. He et al., "Obstacle detection in dangerous railway track areas by a convolutional neural network," *Meas. Sci. Technol.*, vol. 32, no. 10, Oct. 2021, Art. no. 105401.
- [12] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Rich feature hierarchies for accurate object detection and semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 580–587.
- [13] R. Girshick, "Fast R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Santiago, Chile, Dec. 2015, pp. 1440–1448.
- [14] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.
- [15] P. Sermanet, D. Eigen, X. Zhang, M. Mathieu, R. Fergus, and Y. LeCun, "OverFeat: Integrated recognition, localization and detection using convolutional networks," 2013, *arXiv:1312.6229*.

- [16] W. Liu et al., "SSD: Single shot multibox detector," in *Proc. ECCV*, Oct. 2016, pp. 21–37.
- [17] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 779–788.
- [18] W. Liu, A. Rabinovich, and A. C. Berg, "ParseNet: Looking wider to see better," in *Proc. Int. Conf. Learn. Represent.*, 2016, pp. 1–11.
- [19] H. Noh, S. Hong, and B. Han, "Learning deconvolution network for semantic segmentation," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1520–1528.
- [20] V. Badrinarayanan, A. Kendall, and R. Cipolla, "SegNet: A deep convolutional encoder–decoder architecture for image segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 12, pp. 2481–2495, Dec. 2017.
- [21] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. 18th Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, vol. 9351. Cham, Switzerland: Springer, 2015, pp. 234–241.
- [22] L. Jiao et al., "A survey of deep learning-based object detection," *IEEE Access*, vol. 7, pp. 128837–128868, 2019.
- [23] F. Guo, Y. Qian, Y. Wu, Z. Leng, and H. Yu, "Automatic railroad track components inspection using real-time instance segmentation," *Comput.-Aided Civil Infrastruct. Eng.*, vol. 36, no. 3, pp. 362–377, Mar. 2021.
- [24] Simonyan, "Very deep convolutional networks for large-scale image recognition," in *Proc. Int. Conf. Learn. Represent.*, 2015, pp. 1–16.
- [25] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2016, pp. 770–778.
- [26] T. Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 2117–2125.
- [27] S. Liu, L. Qi, H. Qin, J. Shi, and J. Jia, "Path aggregation network for instance segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 8759–8768.
- [28] M. Tan, R. Pang, and Q. V. Le, "EfficientDet: Scalable and efficient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 10778–10787.
- [29] M. Hong, S. Li, Y. Yang, F. Zhu, Q. Zhao, and L. Lu, "SSPNet: Scale selection pyramid network for tiny person detection from UAV images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art. no. 8018505.
- [30] S. Woo, S. Hwang, and I. S. Kweon, "StairNet: Top-down semantic aggregation for accurate one shot detection," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Mar. 2018, pp. 1093–1102.
- [31] Z. Liu, G. Gao, L. Sun, and L. Fang, "IPG-Net: Image pyramid guidance network for small object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2020, pp. 4422–4430.
- [32] H. Xu, L. Yao, Z. Li, X. Liang, and W. Zhang, "Auto-FPN: Automatic network architecture adaptation for object detection beyond classification," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6648–6657.
- [33] G. Ghiasi, T.-Y. Lin, and Q. V. Le, "NAS-FPN: Learning scalable feature pyramid architecture for object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 7029–7038.
- [34] S. Woo, J. Park, J. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 3–19.
- [35] X. Li, Z. Zhong, J. Wu, Y. Yang, Z. Lin, and H. Liu, "Expectation-maximization attention networks for semantic segmentation," in *Proc. IEEE Int. Conf. Comput. Vis.*, Jun. 2019, pp. 9166–9175.
- [36] Z. Huang, X. Wang, L. Huang, C. Huang, Y. Wei, and W. Liu, "CCNet: Criss-cross attention for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 603–612.
- [37] L. Zhu, X. Wang, Z. Ke, W. Zhang, and R. Lau, "BiFormer: Vision transformer with bi-level routing attention," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 10323–10333.
- [38] Y. Liu and J. Tian, "Probabilistic attention map: A probabilistic attention mechanism for convolutional neural networks," *Sensors*, vol. 24, no. 24, p. 8187, Dec. 2024.
- [39] X. He, J. Liu, W. Wang, and H. Lu, "An efficient sampling-based attention network for semantic segmentation," *IEEE Trans. Image Process.*, vol. 31, pp. 2850–2863, 2022.
- [40] D. Heo and H. Choi, "Generalized probabilistic attention mechanism in transformers," 2024, *arXiv:2410.15578*.
- [41] W. Zhangyu, Y. Guizhen, W. Xinkai, L. Haoran, and L. Da, "A camera and LiDAR data fusion method for railway object detection," *IEEE Sensors J.*, vol. 21, no. 12, pp. 13442–13454, Jun. 2021.
- [42] R. Tagiew et al., "Onboard sensor systems for automatic train operation," in *Proc. Eur. Dependable Comput. Conf.*, 2022, pp. 139–150.
- [43] D. B. Kusumawardhana, B. R. Trilaksono, and H. Abdurrohman, "Panoptic segmentation datasets for rail-based autonomous vehicle under mixed-traffic scenario," in *Proc. 12th Int. Conf. Syst. Eng. Technol. (ICSET)*, Oct. 2022, pp. 19–24.
- [44] J. Grandio, B. Riveiro, M. Soilán, and P. Arias, "Point cloud semantic segmentation of complex railway environments using deep learning," *Autom. Construction*, vol. 141, Sep. 2022, Art. no. 104425.
- [45] W. Lu, Q. Wang, J. Ding, W. Niu, and J. Sheng, "Rail track area environment perception based on radar target grid," in *Proc. 3rd Int. Conf. Electron. Commun. Inf. Technol. (CECIT)*, Dec. 2022, pp. 236–241.
- [46] A. Berg, K. Örfjäll, J. Ahlberg, and M. Felsberg, "Detecting rails and obstacles using a train-mounted thermal camera," in *Proc. SCIA*, 2015, pp. 492–503.
- [47] A. O. Lebedev, V. V. Vasilev, B. N. Novgorodov, and A. G. Paulish, "Computer vision controlling an autonomous unmanned aerial vehicle flight over a railway," in *Proc. 1st Int. Conf. Problems Informat., Electron., Radio Eng. (PIERE)*, Novosibirsk, Russia, Dec. 2020, pp. 210–213.
- [48] L. Tong et al., "Fully decoupled residual ConvNet for real-time railway scene parsing of UAV aerial images," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 9, pp. 14806–14819, Sep. 2022.
- [49] M. Salehi and Y. Wang, "Rail network detection from aerial imagery using deep learning," *Comput. Sci., Math.*, 2017. [Online]. Available: <https://cs231n.stanford.edu/reports/2017/posters/564.pdf>
- [50] R. E. Patferson and N. A. M. Mackay, "A guided radar system for obstacle detection," *IEEE Trans. Instrum. Meas.*, vol. IM-26, pp. 137–143, 1977.
- [51] S. Zheng, Z. Wu, Y. Xu, and Z. Wei, "Intrusion detection of foreign objects in overhead power system for preventive maintenance in high-speed railway catenary inspection," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–12, 2022.
- [52] D. Mikhaylov et al., "Toward the future generation of railway localization exploiting RTK and GNSS," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–10, 2023.
- [53] S.-Q. Wang, J. Liu, B.-G. Cai, J. Wang, and D.-B. Lu, "BOSVDD-based GNSS spoofing detection for rail vehicle positioning," *IEEE Trans. Instrum. Meas.*, vol. 74, pp. 1–18, 2025.
- [54] H. Yang, X. Li, Y. Guo, and L. Jia, "Discretization–filtering–reconstruction: Railway detection in images for navigation of inspection UAV," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–13, 2022.
- [55] M. Li, G. Xiong, P. Ye, G. Liu, and F. Zhu, "Model with leader-follower backbone and bifurcation fusion for UAV traffic object detection," *IEEE Trans. Instrum. Meas.*, vol. 74, pp. 1–12, 2025.
- [56] J. Zhao, Y. Chen, Z. Zhou, J. Zhao, S. Wang, and X. Chen, "Multiship speed measurement method based on machine vision and drone images," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–12, 2023.
- [57] Z. Liu et al., "Calibration of a single-point laser range finder and a camera," *Opt. Quantum Electron.*, vol. 50, no. 12, p. 447, Dec. 2018.
- [58] J. Gleichauf, J. Vollet, C. Pfitzner, P. Koch, and S. May, "Sensor fusion approach for an autonomous shunting locomotive," in *Proc. Int. Conf. Inform. Control, Autom. Robot.*, May 2019, pp. 603–624.
- [59] T. Ye, Z. Zhang, X. Zhang, and F. Zhou, "Autonomous railway traffic object detection using feature-enhanced single-shot detector," *IEEE Access*, vol. 8, pp. 145182–145193, 2020.
- [60] M. Ukai, B. T. Nassu, N. Nagamine, M. Watanabe, and T. Inaba, "Obstacle detection on railway track by fusing radar and image sensor," in *Proc. 9th World Congr. Railway Res. (WCRR)*, Lille, France, 2011.
- [61] F. Kruse, S. Milch, and H. Rohling, "Multi sensor system for obstacle detection in train applications," in *Proc. IEEE Tr.*, Jun. 2003, pp. 42–46.
- [62] M. Ruder, N. Mohler, and F. Ahmed, "An obstacle detection system for automated trains," in *Proc. 6th IEEE Intell. Vehicles Symp.*, Jun. 2003, pp. 180–185.
- [63] T. Ye, B. Wang, P. Song, and J. Li, "Automatic railway traffic object detection system using feature fusion refine neural network under shunting mode," *Sensors*, vol. 18, no. 6, p. 1916, Jun. 2018.
- [64] T. Ye, X. Zhang, Y. Zhang, and J. Liu, "Railway traffic object detection using differential feature fusion convolution neural network," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 3, pp. 1375–1387, Mar. 2021.

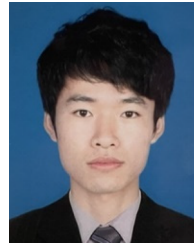
- [65] J. Weichselbaum, C. Zinner, O. Gebauer, and W. Pree, "Accurate 3D-vision-based obstacle detection for an autonomous train," *Comput. Ind.*, vol. 64, no. 9, pp. 1209–1220, Dec. 2013.
- [66] S. Mockel, F. Scherer, and P. F. Schuster, "Multi-sensor obstacle detection on railway tracks," in *Proc. IEEE IV Intell. Vehicles Symp.*, Jun. 2003, pp. 42–46.
- [67] J. A. Uribe, L. Fonseca, and J. F. Vargas, "Video based system for railroad collision warning," in *Proc. IEEE Int. Carnahan Conf. Secur. Technol.*, Oct. 2012, pp. 280–285.
- [68] Y. Wang, L. Wang, Y. H. Hu, and J. Qiu, "RailNet: A segmentation network for railroad detection," *IEEE Access*, vol. 7, pp. 143772–143779, 2019.
- [69] R. Ross, "Vision-based track estimation and turnout detection using recursive estimation," in *Proc. 13th Int. IEEE Conf. Intell. Transp. Syst.*, Sep. 2010, pp. 1330–1335.
- [70] R. Ross, "Track and turnout detection in video-signals using probabilistic spline curves," in *Proc. ITSC*, vol. 38, Sep. 2012, pp. 294–299.
- [71] Z. Qi, Y. Tian, and Y. Shi, "Efficient railway tracks detection and turnouts recognition method using HOG features," *Neural Comput. Appl.*, vol. 23, no. 1, pp. 245–254, Jul. 2013.
- [72] R. Nakasone, N. Nagamine, M. Ukai, H. Mukojima, D. Deguchi, and H. Murase, "Frontal obstacle detection using background subtraction and frame registration," *Quart. Rep. RTRI*, vol. 58, no. 4, pp. 298–302, 2017.
- [73] J. Wohlfeil, "Vision based rail track and switch recognition for self-localization of trains in a rail network," in *Proc. IEEE Intell. Veh. Symp.*, Jun. 2011, pp. 1023–1028.
- [74] B. T. Nassu and M. Ukai, "A vision-based approach for rail extraction and its application in a camera Pan-Tilt control system," *IEEE Trans. Intell. Transp. Syst.*, vol. 13, no. 4, pp. 1763–1771, Dec. 2012.
- [75] B. T. Nassu and M. Ukai, "Rail extraction for driver support in railways," in *Proc. IEEE Intell. Vehicles Symp. (IV)*, Jun. 2011, pp. 83–88.
- [76] S. Athira, "Image processing based real time obstacle detection and alert system for trains," in *Proc. 3rd Int. Conf. Electron., Commun. Aerosp. Technol. (ICECA)*, Jun. 2019, pp. 740–745.
- [77] M. Ukai, "Obstacle detection with a sequence of ultra telephoto camera images," in *Proc. Comput. Railways*, May 2004, pp. 1003–1012.
- [78] M. A. Haseeb, J. Guan, D. Ristic-Durrant, and A. Gräser, "DisNet: A novel method for distance estimation from monocular camera," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Oct. 2018, pp. 1–6.
- [79] D. Ristić-Durrant, M. Haseeb, M. Franke, M. Banić, M. Simonović, and D. Stamenković, "Artificial intelligence for obstacle detection in railways: Project SMART and beyond," in *Proc. Eur. Dependable Comput. Conf.*, 2020, pp. 44–55.
- [80] M. A. Haseeb, D. Ristić-Durrant, and A. Gräser, "Long-range obstacle detection from a monocular camera," in *Proc. ACM Comput. Sci. Cars Symp. (CSCS)*, 2018, pp. 1–3.
- [81] H. M. Abdul, R.-D. Danijela, G. Axel, B. Milan, and S. Dušan, "Multi-DisNet: Machine learning-based object distance estimation from multiple cameras," in *Proc. Int. Conf. Comput. Vis. Syst.*, 2019, pp. 457–469.
- [82] A. Chernov, M. Butakova, A. Guda, and P. Shevchuk, "Development of intelligent obstacle detection system on railway tracks for yard locomotives using CNN," in *Proc. Eur. Dependable Comput. Conf.*, 2020, pp. 33–43.
- [83] O. Zendel, M. Murschitz, M. Zeilinger, D. Steininger, S. Abbasi, and C. Belezna, "RailSem19: A dataset for semantic rail scene understanding," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2019, pp. 1221–1229.
- [84] T. Toprak, B. Belenlioglu, B. Aydin, C. Guzelis, and M. A. Selver, "Conditional weighted ensemble of transferred models for camera based onboard pedestrian detection in railway driver support systems," *IEEE Trans. Veh. Technol.*, vol. 69, no. 5, pp. 5041–5054, May 2020.
- [85] M. A. Haddad, A. Mahtani, S. Ambellouis, J. Boonaert, and H. Wannous, "Application of rail segmentation in the monitoring of autonomous train's frontal environment," in *Proc. Int. Conf. Pattern Recognit. Artif. Intell.*, 2022, pp. 185–197.
- [86] M. Wallner, D. Steininger, V. Widhalm, M. Schörghuber, and C. Belezna, "RGB-D railway platform monitoring and scene understanding for enhanced passenger safety," in *Proc. Int. Conf. Pattern Recognit.*, 2021, pp. 656–671.
- [87] R. Tagiew et al., "OSDaR23: Open sensor data for rail 2023," in *Proc. 8th Int. Conf. Robot. Autom. Eng. (ICRAE)*, Nov. 2023, pp. 270–276.
- [88] R. Tagiew et al., "Sensor system for development of perception systems for ATO," *Discover Artif. Intell.*, vol. 3, no. 1, p. 22, Jun. 2023.
- [89] Y. Chen, N. Zhu, Q. Wu, C. Wu, W. Niu, and Y. Wang, "MRSI: A multimodal proximity remote sensing data set for environment perception in rail transit," *Int. J. Intell. Syst.*, vol. 37, no. 9, pp. 5530–5556, Sep. 2022.
- [90] X. Li and X. Peng, "Rail detection: An efficient row-based network and a new benchmark," in *Proc. 30th ACM Int. Conf. Multimedia*, Oct. 2022, pp. 6455–6463.
- [91] B. Qiu et al., "WHU-railway3D: A diverse dataset and benchmark for railway point cloud semantic segmentation," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 12, pp. 20900–20916, Dec. 2024.
- [92] A. Kharroubi, Z. Ballouch, R. Hajji, A. Yarroudh, and R. Billen, "Multi-context point cloud dataset and machine learning for railway semantic segmentation," *Infrastructures*, vol. 9, no. 4, p. 71, Apr. 2024.
- [93] R. Tagiew, I. Wunderlich, M. Sastuba, K. Göller, and S. Seitz, "RailGoerl24: Gölitz rail test center CV dataset 2024," 2025, *arXiv:2504.00204*.
- [94] C. Gebler and R. Pohle-Fröhlich, "Intrusion detection at railway tunnel entrances using dynamic vision sensors," in *Proc. ICPRAM*, 2024, pp. 902–909.
- [95] M. Franke, V. Gopinath, D. Ristić-Durrant, and K. Michels, "Object-level data augmentation for deep learning-based obstacle detection in railways," *Appl. Sci.*, vol. 12, no. 20, p. 10625, Oct. 2022.
- [96] A. Zouaoui, A. Mahtani, M. A. Haddad, S. Ambellouis, J. Boonaert, and H. Wannous, "RailSet: A unique dataset for railway anomaly detection," in *Proc. IEEE 5th Int. Conf. Image Process. Appl. Syst. (IPAS)*, Dec. 2022, pp. 1–6.
- [97] Q. Guo and J. Rambach, "SynRailObs: A synthetic dataset for obstacle detection in railway scenarios," 2025, *arXiv:2505.10784*.
- [98] J. Ganter, S. Löffler, R. Metzger, K. Ubling, and C. Müller, "Investigating semantic augmentation in virtual environments for image segmentation using convolutional neural networks," *J. Imag.*, vol. 7, no. 8, p. 146, Aug. 2021.
- [99] M. Sevi and I. Aydin, "Rail tracking and detection with drone in gazebo environment," in *Proc. Int. Conf. Decis. Aid Sci. Appl. (DASA)*, Chiangrai, Thailand, Mar. 2022, pp. 1450–1454.
- [100] G. D'Amico et al., "TrainSim: A railway simulation framework for LiDAR and camera dataset generation," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 12, pp. 15006–15017, Dec. 2023.
- [101] S. Minaeian, J. Liu, and Y.-J. Son, "Effective and efficient detection of moving targets from a UAV's camera," *IEEE Trans. Intell. Transp. Syst.*, vol. 19, no. 2, pp. 497–506, Feb. 2018.
- [102] M. Karakose, O. Yaman, and E. Akin, "Detection of rail switch passages through image processing on railway line and use of condition-monitoring approach," in *Proc. Int. Conf. Adv. Technol. Sci.*, 2016, pp. 95–105.
- [103] M. Karakose, O. Yaman, M. Baygin, K. Murat, and E. Akin, "A new computer vision based method for rail track detection and fault diagnosis in railways," *Int. J. Mech. Eng. Robot. Res.*, vol. 6, no. 1, pp. 22–27, 2017.
- [104] H. Wang, X. Zhang, L. Damiani, P. Giribone, R. Revetria, and G. Ronchetti, "Transportation safety improvements through video analysis: An application of obstacles and collision detection applied to railways and roads," in *Proc. Int. MultiConf. Engineers Comput. Scientists*, 2018, pp. 1–15.
- [105] M. Ukai, B. T. Nassu, N. Nagamine, M. Watanabe, and T. Inaba, "Obstacle detection on railway track by fusing radar and image sensor," in *Proc. 9th World Congr. Railway Res. (WCRR)*, 2011, pp. 22–26.
- [106] M. Bertozzi, A. Broggi, and A. Fascioli, "Stereo inverse perspective mapping: Theory and applications," *Image Vis. Comput.*, vol. 16, no. 8, pp. 585–590, Jun. 1998.
- [107] F. Maire and A. Bigdeli, "Obstacle-free range determination for rail track maintenance vehicles," in *Proc. 11th Int. Conf. Control Autom. Robot. Vis.*, Dec. 2010, pp. 2172–2178.
- [108] Z. Wang et al., "An inverse projective mapping-based approach for robust rail track extraction," in *Proc. 8th Int. Congr. Image Signal Process. (CISP)*, Oct. 2015, pp. 888–893.
- [109] Z. Wang et al., "Geometry constraints-based visual rail track extraction," in *Proc. 12th World Congr. Intell. Control Autom. (WCICA)*, Jun. 2016, pp. 993–998.
- [110] S. Yang, G. Yu, Z. Wang, B. Zhou, P. Chen, and Q. Zhang, "A topology guided method for rail-track detection," *IEEE Trans. Veh. Technol.*, vol. 71, no. 2, pp. 1426–1438, Feb. 2022.
- [111] J. Lin and J. Peng, "Adaptive inverse perspective mapping transformation method for ballasted railway based on differential edge detection and improved perspective mapping model," *Digit. Signal Process.*, vol. 135, Apr. 2023, Art. no. 103944.
- [112] M. Gschwandtner, W. Pree, and A. Uhl, "Track detection for autonomous trains," in *Proc. Int. Symp. Vis. Comput.*, 2010, pp. 19–28.

- [113] F. Auger, M. Hilaret, J. M. Guerrero, E. Monmasson, T. Orlowska-Kowalska, and S. Katsura, "Industrial applications of the Kalman filter: A review," *IEEE Trans. Ind. Electron.*, vol. 60, no. 12, pp. 5458–5471, Dec. 2013.
- [114] S.-C. Oh, G.-D. Kim, W.-T. Jeong, and Y.-T. Park, "Vision-based object detection for passenger's safety in railway platform," in *Proc. Int. Conf. Control, Autom. Syst.*, Oct. 2008, pp. 2134–2137.
- [115] M. A. Selver, E. Er, B. Belenlioglu, and Y. Soyaslan, "Camera based driver support system for rail extraction using 2-D Gabor wavelet decompositions and morphological analysis," in *Proc. IEEE Int. Conf. Intell. Rail Transp. (ICIRT)*, Aug. 2016, pp. 270–275.
- [116] M. A. Selver, E. Y. Zoral, B. Belenlioglu, and S. Dogan, "Predictive modeling for monocular vision based rail track extraction," in *Proc. IEEE CISP-BMEI*, Oct. 2017, pp. 1–6.
- [117] H. Mukojima et al., "Moving camera background-subtraction for obstacle detection on railway tracks," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2016, pp. 3967–3971.
- [118] M. Yu, P. Yang, and S. Wei, "Railway obstacle detection algorithm using neural network," *AIP Conf. Proc.*, vol. 1967, pp. –1, May 2018, Art. no. 040017.
- [119] R. Kapoor, R. Goel, and A. Sharma, "Deep learning based object and railway track recognition using train mounted thermal imaging system," *J. Comput. Theor. Nanoscience*, vol. 17, no. 11, pp. 5062–5071, Nov. 2020.
- [120] K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.
- [121] D. He et al., "Improved mask R-CNN for obstacle detection of rail transit," *Measurement*, vol. 190, Feb. 2022, Art. no. 110728.
- [122] R. Liu et al., "Intercity rail transit platform anomaly detection using door tracking-based key frame extraction and AnoDet network," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–16, 2023.
- [123] Y. Xu, C. Gao, L. Yuan, S. Tang, and G. Wei, "Real-time obstacle detection over rails using deep convolutional neural network," in *Proc. IEEE Intell. Transp. Syst. Conf. (ITSC)*, Auckland, New Zealand, Oct. 2019, pp. 1007–1012.
- [124] D. He, Z. Zou, Y. Chen, B. Liu, X. Yao, and S. Shan, "Obstacle detection of rail transit based on deep learning," *Measurement*, vol. 176, May 2021, Art. no. 109241.
- [125] Y. Yang, "High-speed railway clearance intrusion detection using feature fusion enhancement neural network," Dept. Electron. Control Eng., School Mech., Beijing Jiaotong Univ., Beijing, China, Tech. Rep., 2021.
- [126] R. Wang and X. Yu, "Research on orbital foreign object intrusion detection algorithm based on improved YOLOv5," *Yunnan Daxue Xuebao. Ziran Kexue Ban*, vol. 45, no. 4, p. 799, 2023.
- [127] Y. Liu et al., "CNN and binocular vision-based target detection and ranging framework of intelligent railway system," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–11, 2024.
- [128] X. Wang, C. Han, and W. Jin, "BCD-YOLO: A railway perimeter foreign body intrusion detection method based on YOLOv8," *Meas. Sci. Technol.*, vol. 36, no. 5, May 2025, Art. no. 056007.
- [129] X. Ye et al., "Classification and extraction method of hidden dangers along railway lines based on semantic segmentation network," *Int. J. Remote Sens.*, vol. 45, no. 24, pp. 9480–9512, Dec. 2024.
- [130] Z. Zheng et al., "Re-DETR: Research on fast detection technology for railway engineering targets in the dark time domain," *IEEE Access*, vol. 12, pp. 175501–175510, 2024.
- [131] X. Liu, R. Huang, J. Cai, X. Yang, and Z. Xie, "Feasibility analysis of RT-DETR in the context of foreign object intrusion at railway boundaries," in *Proc. Int. Conf. Elect. Inf. Technol. Rail Transp.*, 2023, pp. 538–545.
- [132] S. Chen, "Research and implementation of track abnormal object detection based on multi-sensor information fusion," School Commun. Inf. Eng., Nanjing Univ. Posts Telecommun., Nanjing, China, Tech. Rep., 2023.
- [133] L. Guang, L. Jiang, and Y. Xie, "Research on lightweight model for railway intrusion detection integrating attention mechanism," *Tiedao Xuebao*, vol. 45, no. 5, p. 72, 2023.
- [134] Z. Zhao, J. Kan, J. Liang, T. Ye, and B. Wu, "All-weather intelligent detection system for railway intrusion obstacles based on LGF-Net," *Yiqi Yibiao Xuebao*, vol. 44, no. 9, p. 287, 2023.
- [135] L. Guan, L. Jia, Z. Xie, and C. Yin, "A lightweight framework for obstacle detection in the railway image based on fast region proposal and improved YOLO-tiny network," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–16, 2022.
- [136] Z. Zhao, J. Kang, B. Wu, T. Ye, and J. Liang, "AE-Net: A high accuracy and efficient network for railway obstacle detection based on convolution and transformer," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–14, 2024.
- [137] H. Zhang, J. Huo, Y. Huang, and Q. Liu, "Real-time dynamic visual-inertial SLAM and object tracking based on lightweight deep feature extraction matching," *IEEE Trans. Instrum. Meas.*, vol. 74, pp. 1–22, 2025.
- [138] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2017, pp. 2881–2890.
- [139] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking atrous convolution for semantic image segmentation," 2017, *arXiv:1706.05587*.
- [140] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Apr. 2018.
- [141] K. Grauman and T. Darrell, "The pyramid match kernel: Discriminative classification with sets of image features," in *Proc. 10th IEEE Int. Conf. Comput. Vis. (ICCV)*, vol. 1, Oct. 2005, pp. 1458–1465.
- [142] S. Lazebnik, C. Schmid, and J. Ponce, "Beyond bags of features: Spatial pyramid matching for recognizing natural scene categories," in *Proc. CVPR*, Jun. 2006, pp. 2169–2178.
- [143] L. Sifre and S. Mallat, "Rigid-motion scattering for texture classification," 2014, *arXiv:1403.1687*.
- [144] S. Xie, R. Girshick, P. Dollár, Z. Tu, and K. He, "Aggregated residual transformations for deep neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 1492–1500.
- [145] A. R. Alexandrescu and A. Manole, "A dynamic approach for railway semantic segmentation," *Studia Universitatis Babeş-Bolyai Inform.*, vol. 67, no. 1, pp. 61–76, 2022.
- [146] D. Jha et al., "ResUNet++: An advanced architecture for medical image segmentation," in *Proc. IEEE Int. Symp. Multimedia (ISM)*, Dec. 2019, pp. 225–2255.
- [147] M. Sevi and İ. Aydin, "Improving unet segmentation performance using an ensemble model in images containing railway lines," *Turkish J. Electr. Eng. Comput. Sci.*, vol. 31, no. 4, pp. 739–750, Jul. 2023.
- [148] R. Gasparini et al., "Anomaly detection for vision-based railway inspection," in *Proc. Eur. Dependable Comput. Conf.*, 2020, pp. 56–67.
- [149] R. Gasparini et al., "Anomaly detection, localization and classification for railway inspection," in *Proc. 25th Int. Conf. Pattern Recognit. (ICPR)*, Jan. 2021, pp. 3419–3426.
- [150] K. Jahan, J. P. Umesh, and M. Roth, "Anomaly detection on the rail lines using semantic segmentation and self-supervised learning," in *Proc. IEEE Symp. Ser. Comput. Intell. (SSCI)*, Dec. 2021, pp. 1–7.
- [151] Z. Wang, X. Wu, G. Yu, and M. Li, "Efficient rail area detection using convolutional neural network," *IEEE Access*, vol. 6, pp. 77656–77664, 2018.
- [152] Z. Tao, S. Ren, Y. Shi, X. Wang, and W. Wang, "Accurate and lightweight RailNet for real-time rail line detection," *Electronics*, vol. 10, no. 16, p. 2038, Aug. 2021.
- [153] S. Belyaev, I. Popov, V. Shubnikov, P. Popov, E. Boltenkova, and D. Savchuk, "Railroad semantic segmentation on high-resolution images," in *Proc. IEEE 23rd Int. Conf. Intell. Transp. Syst. (ITSC)*, Sep. 2020, pp. 1–6.
- [154] Y. Zhang, K. Li, G. Zhang, Z. Zhu, and P. Wang, "DFA-UNet: Efficient railroad image segmentation," *Appl. Sci.*, vol. 13, no. 1, p. 662, Jan. 2023.
- [155] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.
- [156] X. Pan, J. Shi, P. Luo, X. Wang, and X. Tang, "Spatial as deep: Spatial CNN for traffic scene understanding," in *Proc. AAAI Conf. Artif. Intell.*, vol. 32, no. 1, Apr. 2018, pp. 1–8.
- [157] H. Li, Q. Zhang, D. Zhao, and Y. Chen, "RailNet: An information aggregation network for rail track segmentation," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Jul. 2020, pp. 1–7.
- [158] L. Lian et al., "Densely multiscale fusion network for lightweight and accurate semantic segmentation of railway scenes," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–11, 2024.
- [159] A. Boussik, W. Ben-Messaoud, S. Niar, and A. Taleb-Ahmed, "Railway obstacle detection using unsupervised learning: An exploratory study," in *Proc. IEEE Intell. Vehicles Symp. (IV)*, Jul. 2021, pp. 660–667.
- [160] Z. Gao, B. Peng, T. Li, and C. Gou, "Generative adversarial networks for road crack image segmentation," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Jul. 2019, pp. 1–8.

- [161] B. Guo, G. Geng, L. Zhu, H. Shi, and Z. Yu, "High-speed railway intruding object image generating with generative adversarial networks," *Sensors*, vol. 19, no. 14, p. 3075, Jul. 2019.
- [162] Z. Zheng and W. Liu, "Research on foreign object intrusion detection algorithm of metro rail based on fast diffusion generation model," *Railway Standard Des.*, vol. 68, no. 6, pp. 191–200, 2024.
- [163] X. Deng, "Unsupervised track abnormal object detection," School Comput. Sci. Technol., Beijing Jiaotong Univ., Beijing, China, Tech. Rep., 2021.
- [164] Y. Liu, "Research on orbital foreign object intrusion detection based on image super-resolution reconstruction algorithm," School Automat. Elect. Eng., Lanzhou Jiaotong Univ., Lanzhou, China, Tech. Rep., 2021.
- [165] X. Gong, X. Chen, Z. Zhong, and W. Chen, "Enhanced few-shot learning for intrusion detection in railway video surveillance," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 8, pp. 11301–11313, Aug. 2022.
- [166] T. Ye, Z. Zheng, X. Li, Z. Zhao, and X.-Z. Gao, "An efficient few-shot object detection method for railway intrusion via fine-tune approach and contrastive learning," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–13, 2023.
- [167] T. Zheng et al., "CLRNet: Cross layer refinement network for lane detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 898–907.
- [168] H. Ni, R. Wang, and S. Liu, "Ultra-fast adaptive track detection network," 2024, *arXiv:2405.13538*.
- [169] L. Tong et al., "TriRNet: Real-time rail recognition network for UAV-based railway inspection," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 5, pp. 3927–3943, May 2024.
- [170] X. Li, J. Li, X. Hu, and J. Yang, "Line-CNN: End-to-end traffic line detection with line proposal unit," *IEEE Trans. Intell. Transp. Syst.*, vol. 21, no. 1, pp. 248–258, Jan. 2020.
- [171] B. Zhou, G. Xing, and Y. Liu, "Rail line detection algorithm based on improved CLRNet," *J. Shanghai Jiaotong Univ. Sci.*, vol. 29, pp. 1–12, Jan. 2024.
- [172] Z. Yu, Q. Liu, W. Wang, L. Zhang, and X. Zhao, "DALNet: A rail detection network based on dynamic anchor line," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–14, 2024.
- [173] K. Huang, Y. Shen, J. Chen, L. Wang, S. Wang, and P. Dai, "FRVomono: Feature-based railway visual odometry with monocular camera," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–10, 2023.
- [174] X. Wang and Q. Wang, "Design of railway foreign object intrusion warning system based on multimodal information fusion," *Electron. Technol.*, vol. 51, no. 12, p. 344, 2022.
- [175] I. Ćirić, M. Pavlović, D. Ristić-Durrant, L. Dimitrov, and V. Nikolić, "FuzzyH method for distance estimation in autonomous train operation," *Symmetry*, vol. 17, no. 4, p. 509, Mar. 2025.
- [176] M. Karaduman, "Image processing based obstacle detection with laser measurement in railways," in *Proc. 10th Int. Conf. Electr. Electron. Eng. (ELECO)*, Jan. 2017, pp. 899–903.
- [177] Z. Wang, G. Yu, P. Chen, B. Zhou, and S. Yang, "FarNet: An attention-aggregation network for long-range rail track point cloud segmentation," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 8, pp. 13118–13126, Aug. 2022.
- [178] S. Yang et al., "RailDepth: A self-supervised network for railway depth completion based on a pooling-guidance mechanism," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–13, 2023.
- [179] J. Cao, Y. Li, and S. Du, "Robust artificial intelligence-aided multimodal rail-obstacle detection method by rail track topology reconstruction," *Appl. Sci.*, vol. 14, no. 7, p. 2795, Mar. 2024.
- [180] R. Bai, Z. Wu, and T. Xu, "A lightweight camera and LiDAR fusion framework for railway transit obstacle detection," in *Proc. 3rd Asia Conf. Algorithms, Comput. Mach. Learn.*, Mar. 2024, pp. 303–308.
- [181] R. Kageyama, N. Nagamine, and H. Mukojima, "Train frontal obstacle detection method with camera-LiDAR fusion," *Quart. Rep. RTRI*, vol. 63, no. 3, pp. 181–186, 2022.
- [182] C. Walters and S. Hadfield, "EVReflex: Dense time-to-impact prediction for event-based obstacle avoidance," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Sep. 2021, pp. 1304–1309.
- [183] F. Tschopp et al., "Hough²map—Iterative event-based Hough transform for high-speed railway mapping," *IEEE Robot. Autom. Lett.*, vol. 6, no. 2, pp. 2745–2752, Apr. 2021.
- [184] W. Wei, Y. Cheng, J. He, and X. Zhu, "A review of small object detection based on deep learning," *Neural Comput. Appl.*, vol. 36, no. 12, pp. 6283–6303, 2024.
- [185] C. Li et al., "YOLOv6: A single-stage object detection framework for industrial applications," 2022, *arXiv:2209.02976*.
- [186] S. Liu, D. Huang, and Y. Wang, "Learning spatial fusion for single-shot object detection," 2019, *arXiv:1911.09516*.
- [187] X. Li, W. Wang, X. Hu, and J. Yang, "Selective kernel networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 510–519.
- [188] H. Zhang et al., "ResNest: Split-attention networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jul. 2022, pp. 2735–2745.
- [189] Y. Dai, F. Giesecke, S. Oehmcke, Y. Wu, and K. Barnard, "Attentional feature fusion," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2021, pp. 3560–3569.
- [190] X. Zeng et al., "Crafting GBD-Net for object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 9, pp. 2109–2123, Sep. 2018.
- [191] X. Tang, D. K. Du, Z. He, and J. Liu, "PyramidBox: A context-assisted single shot face detector," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 812–828.
- [192] S. Lee et al., "VPGNet: Vanishing point guided network for lane and road marking detection and recognition," in *Proc. IEEE Int. Conf. Comput. Vis.*, Oct. 2017, pp. 1947–1955.
- [193] G. Chen, K. Chen, L. Zhang, L. Zhang, and A. Knoll, "VCANet: Vanishing-point-guided context-aware network for small road object detection," *Automot. Innov.*, vol. 4, no. 4, pp. 400–412, Nov. 2021.
- [194] Y. Yuan, Z. Xiong, and Q. Wang, "VSSA-NET: Vertical spatial sequence attention network for traffic sign detection," *IEEE Trans. Image Process.*, vol. 28, no. 7, pp. 3423–3434, Jul. 2019.
- [195] D. Yuan, X. Chang, Z. Li, and Z. He, "Learning adaptive spatial-temporal context-aware correlation filters for UAV tracking," *ACM Trans. Multimedia Comput., Commun., Appl.*, vol. 18, no. 3, pp. 1–18, Aug. 2022.
- [196] X. Ni et al., "Supapixel-guided multi-type rail segmentation via contextual information aggregation," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 10, pp. 14004–14018, Oct. 2024.
- [197] A. Creswell, T. White, V. Dumoulin, K. Arulkumaran, B. Sengupta, and A. A. Bharath, "Generative adversarial networks: An overview," *IEEE Signal Process. Mag.*, vol. 35, no. 1, pp. 53–65, Jan. 2018.
- [198] J. Li, X. Liang, Y. Wei, T. Xu, J. Feng, and S. Yan, "Perceptual generative adversarial networks for small object detection," in *Proc. Conf. Comput. Vis. Pattern Recognit.*, Jun. 2017, pp. 1951–1959.
- [199] J. Noh, W. Bae, W. Lee, J. Seo, and G. Kim, "Better to follow, follow to be better: Towards precise supervision of feature super-resolution for small object detection," in *Proc. IEEE Int. Conf. Comput. Vis.*, Oct. 2019, pp. 9724–9733.
- [200] J. Liu et al., "Inception convolution with efficient dilation search," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 11486–11495.
- [201] Y. Yuan, X. Zheng, and X. Lu, "Hyperspectral image superresolution by transfer learning," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 5, pp. 1963–1974, May 2017.
- [202] L. Courtrai, M.-T. Pham, and S. Lefèvre, "Small object detection in remote sensing images based on super-resolution with auxiliary generative adversarial networks," *Remote Sens.*, vol. 12, no. 19, p. 3152, Sep. 2020.
- [203] J. Y. Zhu, T. Park, P. Isola, and A. A. Efros, "Unpaired image-to-image translation using cycle-consistent adversarial networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 81–88.
- [204] I. Gulrajani, F. Ahmed, M. Arjovsky, V. Dumoulin, and A. C. Courville, "Improved training of Wasserstein GANs," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, 2017, pp. 5767–5777.
- [205] B. Lim, S. Son, H. Kim, S. Nah, and K. M. Lee, "Enhanced deep residual networks for single image super-resolution," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jul. 2017, pp. 1132–1140.
- [206] J. Rabbi, N. Ray, M. Schubert, S. Chowdhury, and D. Chao, "Small-object detection in remote sensing images with end-to-end edge-enhanced GAN and object detector network," *Remote Sens.*, vol. 12, no. 9, p. 1432, May 2020.
- [207] K. Jiang, Z. Wang, P. Yi, G. Wang, T. Lu, and J. Jiang, "Edge-enhanced GAN for remote sensing image superresolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5799–5812, Aug. 2019.
- [208] X. Wang et al., "ESRGAN: Enhanced super-resolution generative adversarial networks," in *Proc. ECCV*, 2018, pp. 63–79.
- [209] P. Dhariwal and A. Nichol, "Diffusion models beat GANs on image synthesis," in *Proc. NIPS*, vol. 34, 2021, pp. 8780–8794.
- [210] Z. Zhang et al., "Introducing shape prior module in diffusion model for medical image segmentation," 2023, *arXiv:2309.05929*.

- [211] F. Liu and W. Huang, "ESDiff: A joint model for low-quality retinal image enhancement and vessel segmentation using a diffusion model," *Biomed. Opt. Exp.*, vol. 14, no. 12, pp. 6563–6578, Nov. 2023.
- [212] M. Kisantal, Z. Wojna, J. Murawski, J. Naruniec, and K. Cho, "Augmentation for small object detection," in *Proc. CS IT Conf.*, vol. 9, 2019, pp. 1–15.
- [213] M. Zhao, L. Cheng, X. Yang, P. Feng, L. Liu, and N. Wu, "TBC-Net: A real-time detector for infrared small target detection using semantic constraint," 2019, *arXiv:2001.05852*.
- [214] C. Chen et al., "RRNet: A hybrid detector for object detection in drone-captured images," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshop (ICCVW)*, Oct. 2019, pp. 100–108.
- [215] H. Yu et al., "A foreign object detection method for railway overhead lines based on few-shot learning," in *Proc. Int. Conf. Elect. Inf. Technol. Rail Transp.*, 2024, pp. 128–134.
- [216] C. Chen, K. Li, W. Wei, J. T. Zhou, and Z. Zeng, "Hierarchical graph neural networks for few-shot learning," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 1, pp. 240–252, Jan. 2022.
- [217] K. Li, Y. Zhang, K. Li, and Y. Fu, "Adversarial feature hallucination networks for few-shot learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2020, pp. 13470–13479.
- [218] J. Wang, Y. Zou, P. Lei, R. S. Sherratt, and L. Wang, "Research on recurrent neural network based crack opening prediction of concrete dam," *J. Internet Technol.*, vol. 21, no. 4, pp. 1161–1169, 2020.
- [219] Y. Duan, W. Zheng, X. Lin, J. Lu, and J. Zhou, "Deep adversarial metric learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 2780–2789.
- [220] H. Dong, K. Song, Q. Wang, Y. Yan, and P. Jiang, "Deep metric learning-based for multi-target few-shot pavement distress classification," *IEEE Trans. Ind. Informat.*, vol. 18, no. 3, pp. 1801–1810, Mar. 2022.
- [221] H. Yu, C. Peng, J. Liu, J. Zhang, and L. Liu, "Improved metric-learning-based recognition method for rail surface state with small-sample data," *IEEE Access*, vol. 12, pp. 4985–4996, 2024.
- [222] T. Yang, Y. Liu, Y. Huang, J. Liu, and S. Wang, "Symmetry-driven unsupervised abnormal object detection for railway inspection," *IEEE Trans. Ind. Informat.*, vol. 19, no. 12, pp. 11487–11498, Dec. 2023.
- [223] Y. Xu, Y. Bao, Y. Zhang, and H. Li, "Attribute-based structural damage identification by few-shot meta learning with inter-class knowledge transfer," *Struct. Health Monitor.*, vol. 20, no. 4, pp. 1494–1517, Jul. 2021.
- [224] Y. Liu, X. Wei, P. Lasang, S. Pranata, K. Subramanian, and H. Seow, "Ensemble uncertainty guided road scene anomaly detection: A simple meta-learning approach," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 9, pp. 10754–10765, Sep. 2024.
- [225] S. Kim, D. Hwang, S. Woo, and R. Singh, "Does prior data matter? Exploring joint training in the context of few-shot class-incremental learning," 2025, *arXiv:2503.10003*.
- [226] E. Okon, V. Rachakonda, H. J. Hong, C. Callison-Burch, and J. B. Lipoff, "Natural language processing of Reddit data to evaluate dermatology patient experiences and therapeutics," *J. Amer. Acad. Dermatol.*, vol. 83, no. 3, pp. 803–808, Sep. 2020.
- [227] A. Esteve et al., "Dermatologist-level classification of skin cancer with deep neural networks," *Nature*, vol. 542, no. 7639, pp. 115–118, Feb. 2017.
- [228] A. Coustasse, R. Sarkar, B. Abodunde, B. J. Metzger, and C. M. Slater, "Use of teledermatology to improve dermatological access in rural areas," *Telemedicine E-Health*, vol. 25, no. 11, pp. 1022–1032, Nov. 2019.
- [229] M. W. Tsang and J. S. Resneck, "Even patients with changing moles face long dermatology appointment wait-times: A study of simulated patient calls to dermatologists," *J. Amer. Acad. Dermatol.*, vol. 55, no. 1, pp. 54–58, Jul. 2006.
- [230] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 8748–8763.
- [231] C. Jia et al., "Scaling up visual and vision-language representation learning with noisy text supervision," in *Proc. 38th Int. Conf. Mach. Learn.*, 2021, pp. 4904–4916.
- [232] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-resolution image synthesis with latent diffusion models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 10684–10695.
- [233] B. Wang, Q. Chen, and Z. Wang, "Diffusion-based visual art creation: A survey and new perspectives," 2024, *arXiv:2408.12128*.
- [234] Y. Cong et al., "FLATTEN: Optical FLOW-guided ATTENTION for consistent text-to-video editing," 2023, *arXiv:2310.05922*.
- [235] Y. Xu, B. Huang, X. Luo, K. Bradbury, and J. M. Malof, "SIMPL: Generating synthetic overhead imagery to address custom zero-shot and few-shot detection problems," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 4386–4396, 2022.
- [236] S. R. Richter, V. Vineet, S. Roth, and V. Koltun, "Playing for data: Ground truth from computer games," in *Proc. Eur. Conf. Comput. Vis.*, in Lecture Notes in Computer Science, 2016, pp. 102–118.
- [237] Y. Zhang et al., "Physically-based rendering for indoor scene understanding using convolutional neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 5057–5065.
- [238] J. Tobin, R. Fong, A. Ray, J. Schneider, W. Zaremba, and P. Abbeel, "Domain randomization for transferring deep neural networks from simulation to the real world," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, Sep. 2017, pp. 23–30.
- [239] J. Pan, Q. Peng, S. Zhan, and J. Bai, "Multiscenario-based train headway analysis under virtual coupling system," *J. Adv. Transp.*, vol. 2021, pp. 1–20, Dec. 2021.
- [240] M. Labayen, X. Mendiadua, N. Aginako, and B. Sierra, "Semi-automatic validation and verification framework for CV&AI-enhanced railway signaling and landmark detector," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–13, 2023.
- [241] H. Han, X. Chen, and C. Li, "3D simulation scene optimization model of high-speed railway virtual reality based on artificial neural network," in *Proc. World Conf. Intell. 3D Technol.*, 2023, pp. 165–177.
- [242] X. Liu et al., "A systematic review of digital twin about physical entities, virtual models, twin data, and applications," *Adv. Eng. Informat.*, vol. 55, Jan. 2023, Art. no. 101876.
- [243] G. Zhai, Y. Xu, and B. F. Spencer, "Bidirectional graphics-based digital twin framework for quantifying seismic damage of structures using deep learning networks," *Struct. Health Monitor.*, vol. 24, no. 1, pp. 86–110, 2024.
- [244] V. Hoskere, Y. Narazaki, B. F. Spencer, and M. D. Smith, "Deep learning-based damage detection of miter gates using synthetic imagery from computer graphics," in *Proc. 12th Int. Workshop Struct. Health Monitor.*, Stanford, CA, USA, 2019, pp. 3073–3080.
- [245] A. Tulembayeva, S. Ashirov, E. Makarov, and Z. Seitbattalov, "Digital model of automated unmanned aerial vehicle with edge computing application for railway reconnaissance," *Acta Polytechnica*, vol. 64, no. 1, pp. 59–67, Mar. 2024.
- [246] Y. Narazaki, V. Hoskere, K. Yoshida, B. F. Spencer, and Y. Fujino, "Synthetic environments for vision-based structural condition assessment of Japanese high-speed railway viaducts," *Mech. Syst. Signal Process.*, vol. 160, Nov. 2021, Art. no. 107850.
- [247] Z. Guo, Z. Tan, J. Feng, and J. Zhou, "VesselDiffusion: 3D vascular structure generation based on diffusion model," *IEEE Trans. Med. Imag.*, vol. 44, no. 9, pp. 3845–3857, Sep. 2025.
- [248] S. Meng, W. Chen, and Y. Jiang, "A novel fast railway foreign objects intrusion detection method based on cascaded convolution neural network and knowledge distillation," v1, Res. Square, Durham, NC, USA, 2024.
- [249] A. Evain et al., "Improving the efficiency of 3D monocular object detection and tracking for road and railway smart mobility," *Sensors*, vol. 23, no. 6, p. 3197, Mar. 2023.
- [250] Y. Shen, D. He, Q. Liu, Z. Jin, X. Li, and C. Ren, "Unsupervised intrusion detection for rail transit based on anomaly segmentation," *Signal, Image Video Process.*, vol. 18, no. 2, pp. 1079–1087, Mar. 2024.
- [251] B. Guo, J. Shi, L. Zhu, and Z. Yu, "High-speed railway clearance intrusion detection with improved SSD network," *Appl. Sci.*, vol. 9, no. 15, p. 2981, Jul. 2019.
- [252] Q. Jiang, J. Dai, and T. Rui, "Lightweight object detection algorithm based on hybrid pruning strategy," in *Proc. Int. Conf. Image Vis. Intell. Syst. (ICIVIS)*, 2023, pp. 701–712.
- [253] C. Chenglin, W. Fei, Y. Min, Q. Yong, and B. Yun, "Edge-enabled real-time railway track segmentation," 2024, *arXiv:2401.11492*.
- [254] W. Wang, Y. Zhang, and L. Zhu, "DRF-DRC: Dynamic receptive field and dense residual connections for model compression," *Cognit. Neurodynamics*, vol. 17, no. 6, pp. 1561–1573, Dec. 2023.
- [255] Y. Wang and P. Yu, "A fast intrusion detection method for high-speed railway clearance based on low-cost embedded GPUs," *Sensors*, vol. 21, no. 21, p. 7279, Nov. 2021.
- [256] H. Wang, R. Mo, Y. Chen, W. Lin, M. Xu, and B. Liu, "Pedestrian and vehicle detection based on pruning YOLOv4 with cloud-edge collaboration," *Comput. Model. Eng. Sci.*, vol. 137, no. 2, pp. 2025–2047, 2023.

- [257] H. Liu, K. Fan, Q. Ouyang, and N. Li, "Real-time small drones detection based on pruned YOLOv4," *Sensors*, vol. 21, no. 10, p. 3374, May 2021.
- [258] K. Sarvajcz, L. Ari, and J. Menyhart, "AI on the road: NVIDIA Jetson nano-powered computer vision-based system for real-time pedestrian and priority sign detection," *Appl. Sci.*, vol. 14, no. 4, p. 1440, Feb. 2024.
- [259] G. Carvalho, B. Cabral, V. Pereira, and J. Bernardino, "Edge computing: Current trends, research challenges and future directions," *Computing*, vol. 103, no. 5, pp. 993–1023, May 2021.
- [260] Z. Zhang, P. Chen, Y. Huang, L. Dai, F. Xu, and H. Hu, "Railway obstacle intrusion warning mechanism integrating YOLO-based detection and risk assessment," *J. Ind. Inf. Integr.*, vol. 38, Mar. 2024, Art. no. 100571.
- [261] A. L. Amin, D. Chimba, and K. Hasan, "Integrating AI and edge computing for advanced safety at railroad grade crossings," *J. Rail Transp. Planning Manage.*, vol. 33, Mar. 2025, Art. no. 100501.
- [262] E. Humes, M. Navardi, and T. Mohsenin, "Squeezed edge YOLO: Onboard object detection on edge devices," 2023, *arXiv:2312.11716*.



Wen-Qiang Liu (Member, IEEE) received the B.S. degree in electronic information engineering and the Ph.D. degree in electrical engineering from Southwest Jiaotong University, China, in 2013 and 2021, respectively.

From 2017 to 2019, he was a Visiting Scholar with the Department of Engineering Structures, Delft University of Technology, Delft, The Netherlands. He is currently a Research Fellow with the National Engineering Research Center on Rail Transit Electrification and Automation (Hong Kong Branch), The

Hong Kong Polytechnic University, Hong Kong, China. His research interests include artificial intelligence, computer vision, imaging, signal processing, and their applications in fault diagnosis and maintenance of high-speed railway and maglev infrastructures.



Xin-Yue Xu received the B.S. degree in civil engineering and the M.S. degree in highway and railway engineering from Central South University, Changsha, China, in 2020 and 2023, respectively. She is currently pursuing the Ph.D. degree in civil and environmental engineering with The Hong Kong Polytechnic University, Hong Kong, China.

Her research interests include artificial intelligence, computer vision, imaging, and their applications in railway engineering.



Su-Mei Wang (Member, IEEE) received the B.S. degree in civil engineering from Hebei University, Baoding, China, in 2013, and the Ph.D. degree in bridge and tunnel engineering from Zhejiang University, Hangzhou, China, in 2018.

She is currently a Research Assistant Professor at The Hong Kong Polytechnic University, Hong Kong, China. Her research interests include structural health monitoring, high-speed railway and maglev structural dynamics, interaction of train–rail–bridge system, vision-based deep learning, and monitoring and control in rail engineering.



Yi-Qing Ni (Member, IEEE) received the B.Eng. degree in structural engineering and the M.S. degree in structural dynamics from Zhejiang University, Hangzhou, China, in 1983 and 1986, respectively, and the Ph.D. degree in structural dynamics from The Hong Kong Polytechnic University (PolyU), Hong Kong, China, in 1997.

He is currently a Yim, Mak, Kwok, and Chung Professor of smart structures and the Chair Professor of smart structures and rail transit at PolyU. He is also the Director of the National Engineering Research Center on Rail Transit Electrification and Automation (Hong Kong Branch), PolyU, and the Director of the PolyU-Hangzhou Technology and Innovation Research Institute, Hangzhou, China. His research interests include structural health monitoring, structural dynamics and control, smart materials and structures, sensors and actuators, and high-speed rail and maglev safety.